

Short-time Fourier transform & Spectrogram

by Dr. Nafaa NACEREDDINE

Outlines

- Concept of windowing
- Time-frequency resolution trade-off
- Spectrogram: visualization and interpretation
- Window types and lengths

References

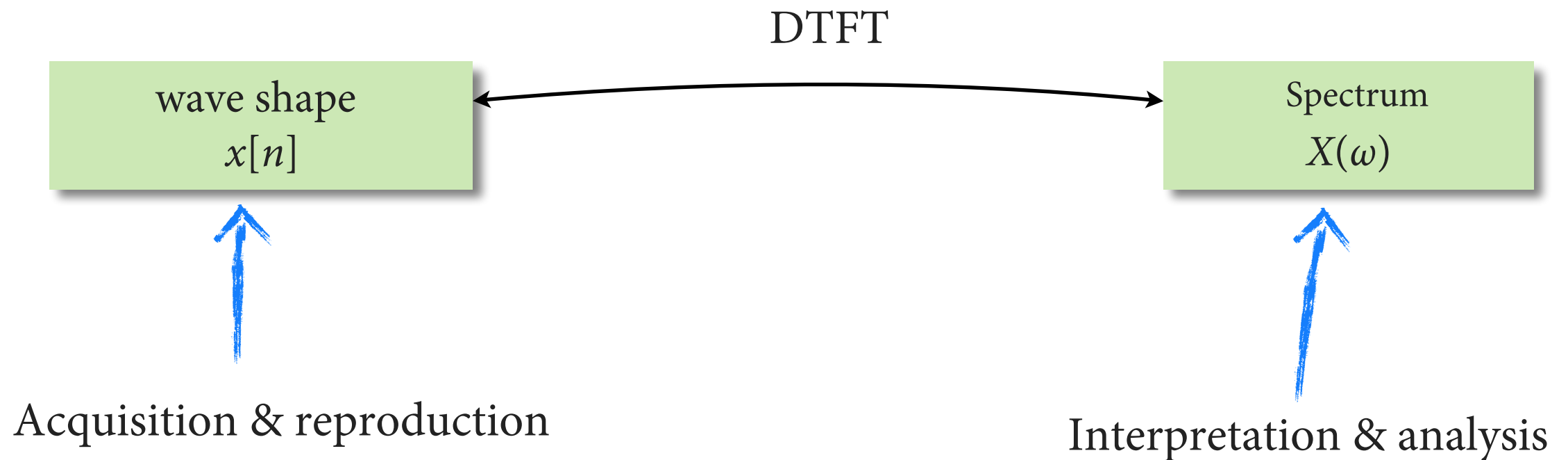
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- A.V.Oppenheim and R.W.Schafer, *Discrete-Time Signal Processing*, Prentice Hall, 1998.
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- B. Champagne et F. Labeau. *Discrete Time Signal Processing*, lecture notes, 2004

Context

Relationships between various forms of speech signals

Recall

wave vs frequency

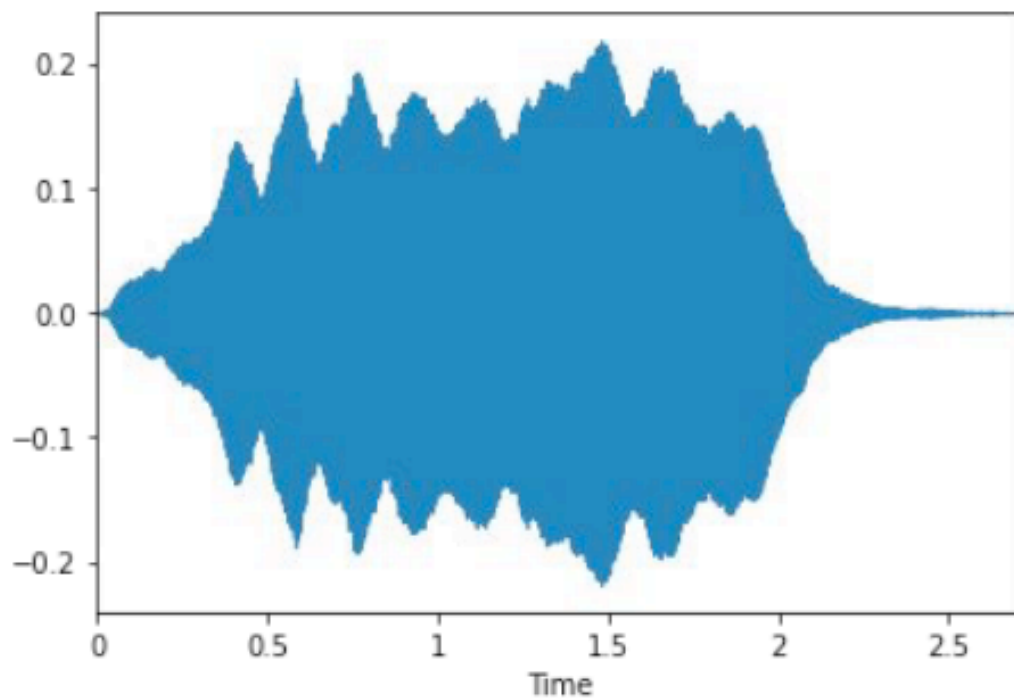


Context

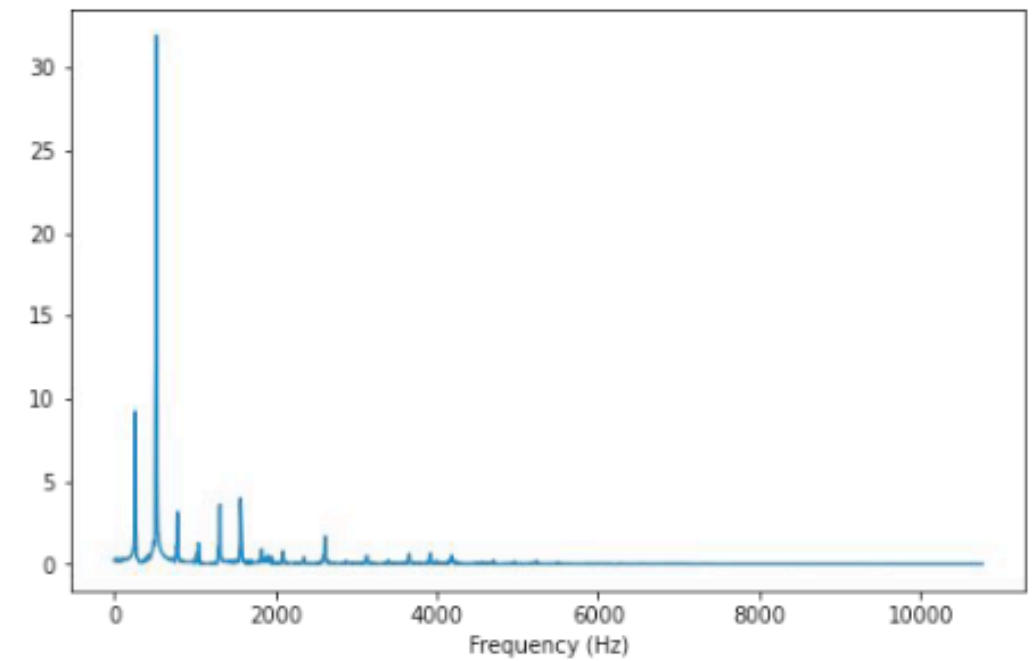
Short-time Fourier transform

Problematic

wave vs frequency



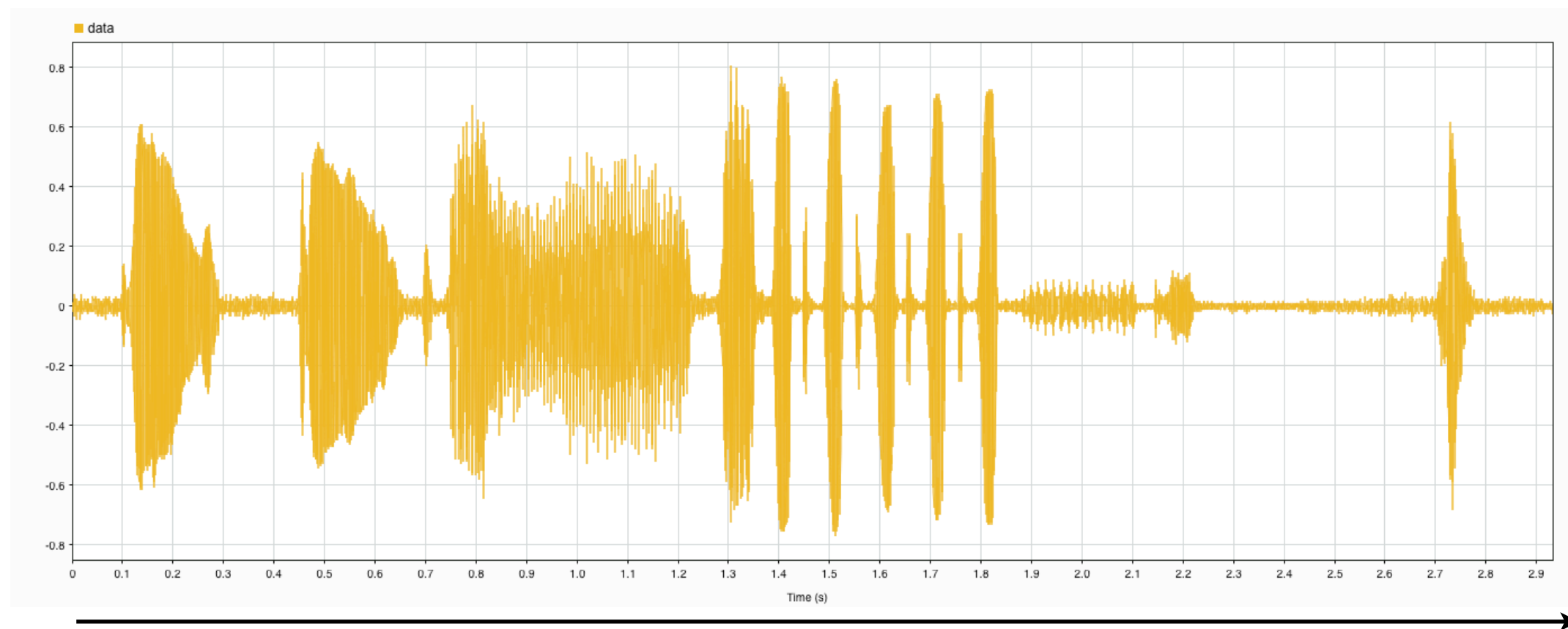
DFT



Context

Short-time Fourier transform

Problematic



Speech Record

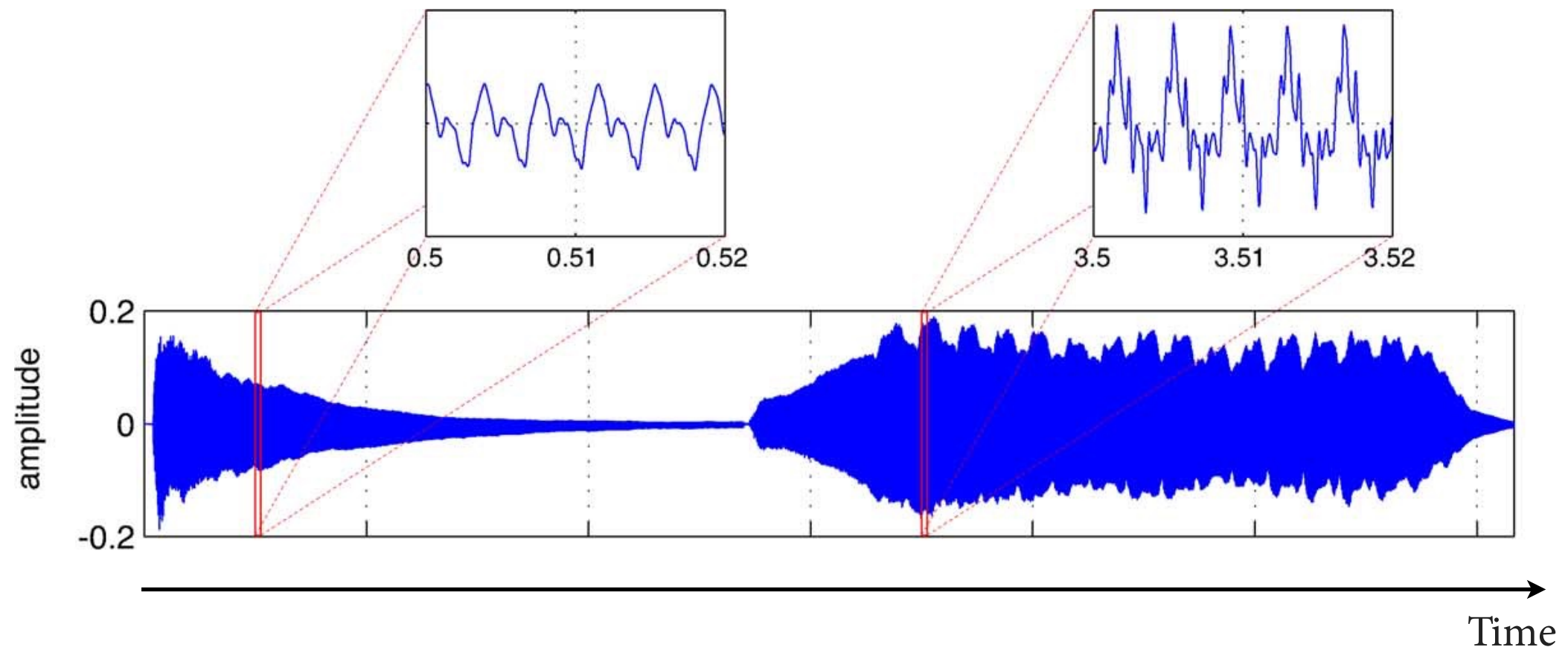
Time

Context

Short-time Fourier transform

Problematic

sound not stationary

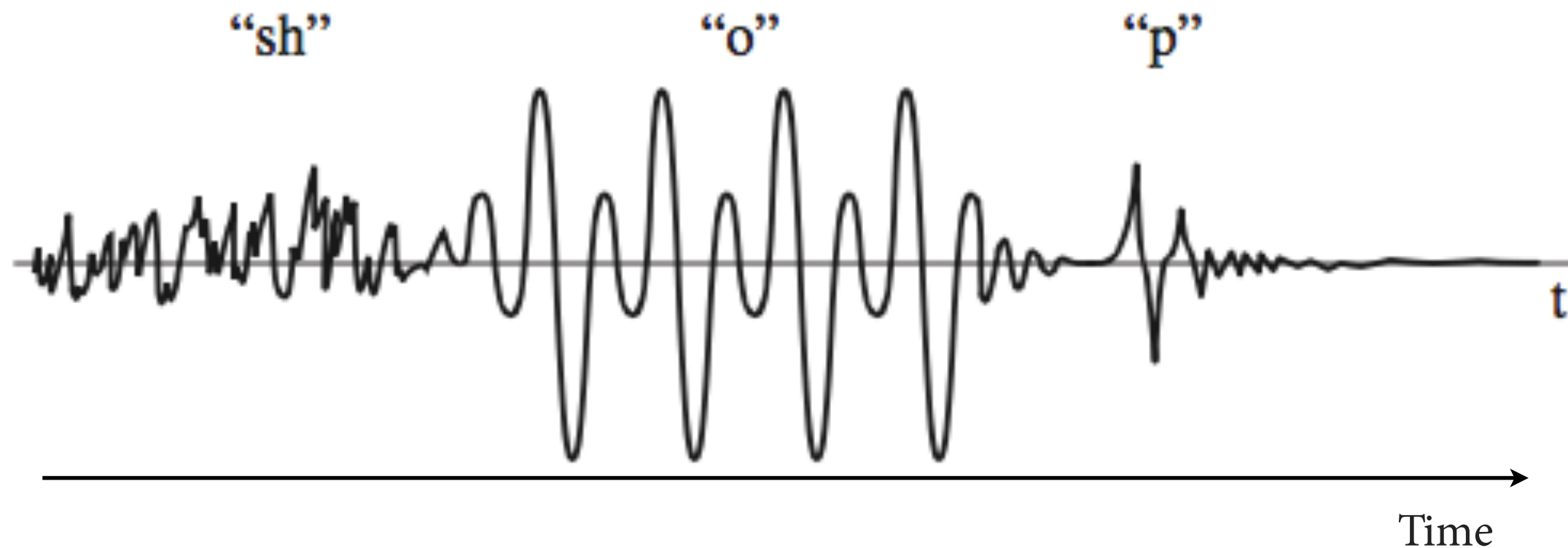


Context

Short-time Fourier transform

Problematic

sound not stationary



Speech signal

Context

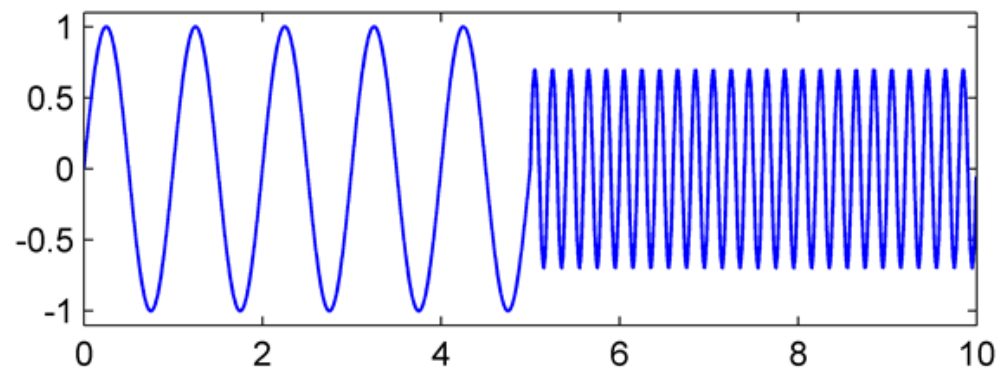
Short-time Fourier transform

Problematic

FT does not conserve the time information

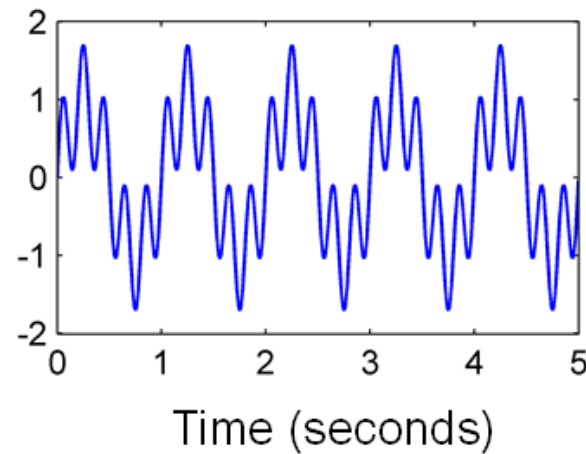
➡ we know what, we don't know when...

Two signals with the same two harmonic waves

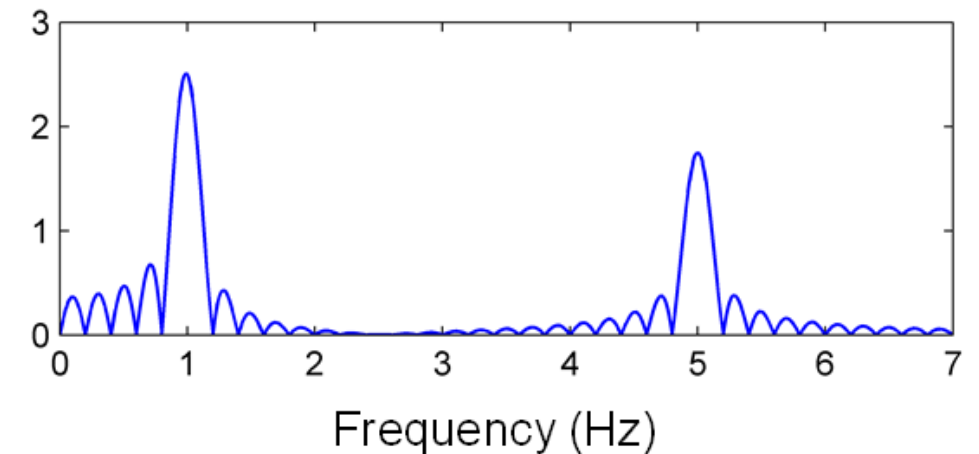
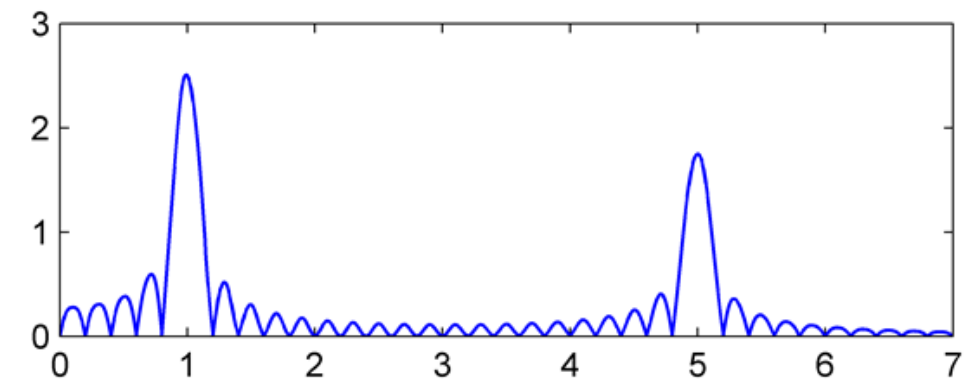


One after
the other

Superposed



Spectral composition is the same in both cases

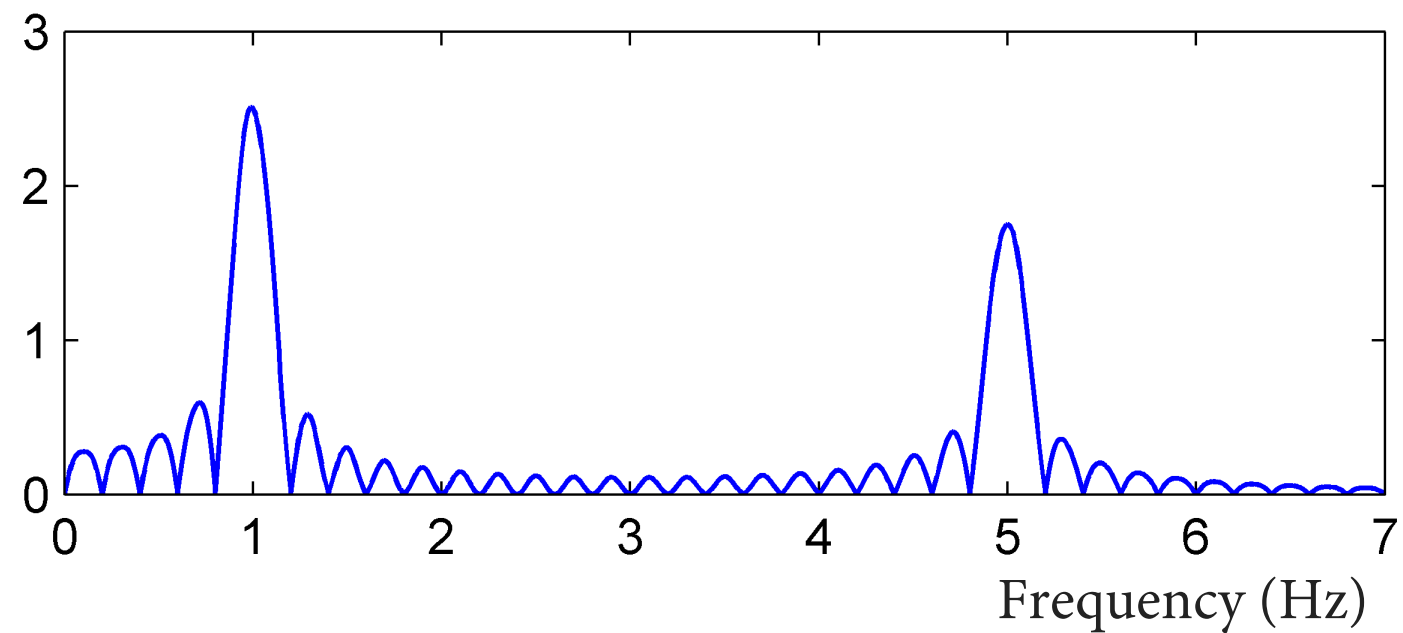
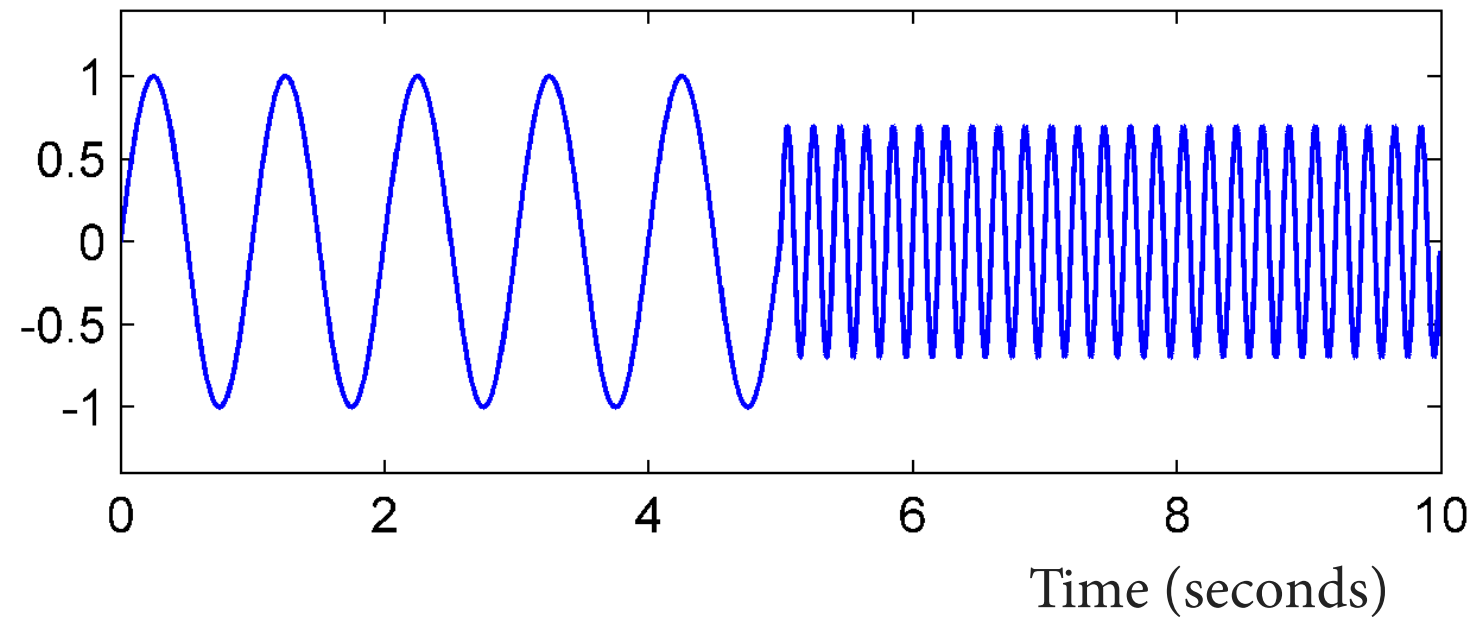


Frequency (Hz)

Context

Short-time Fourier transform

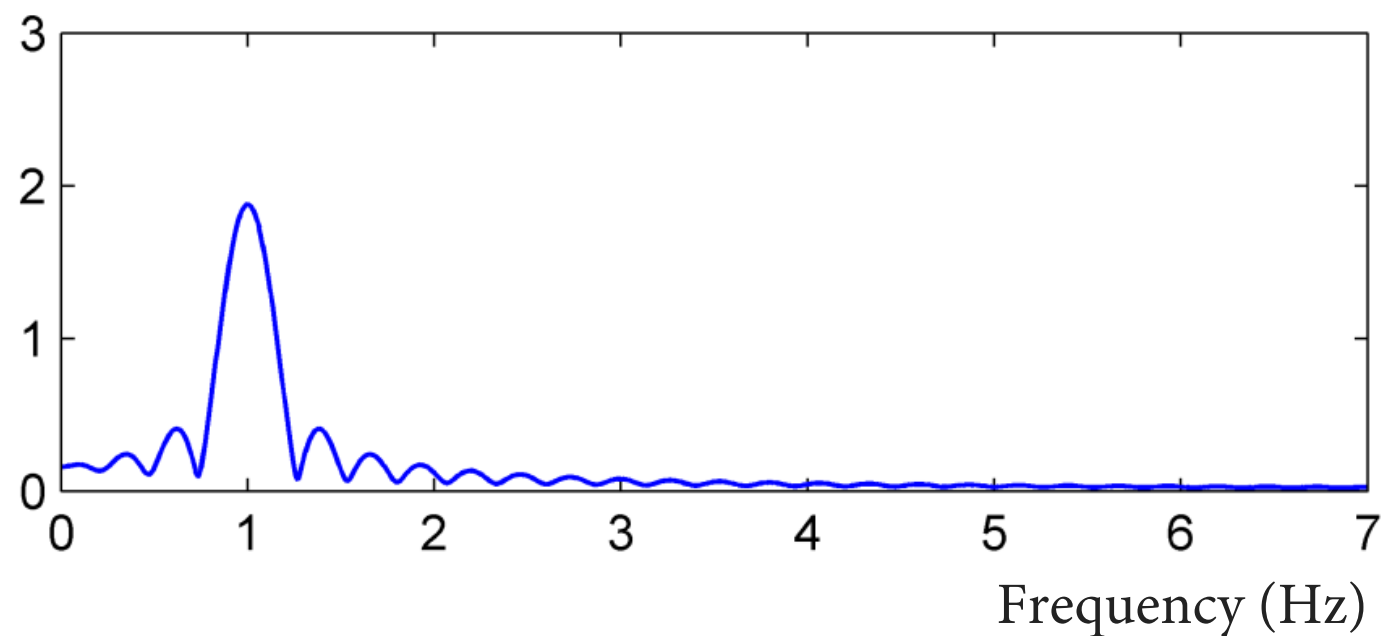
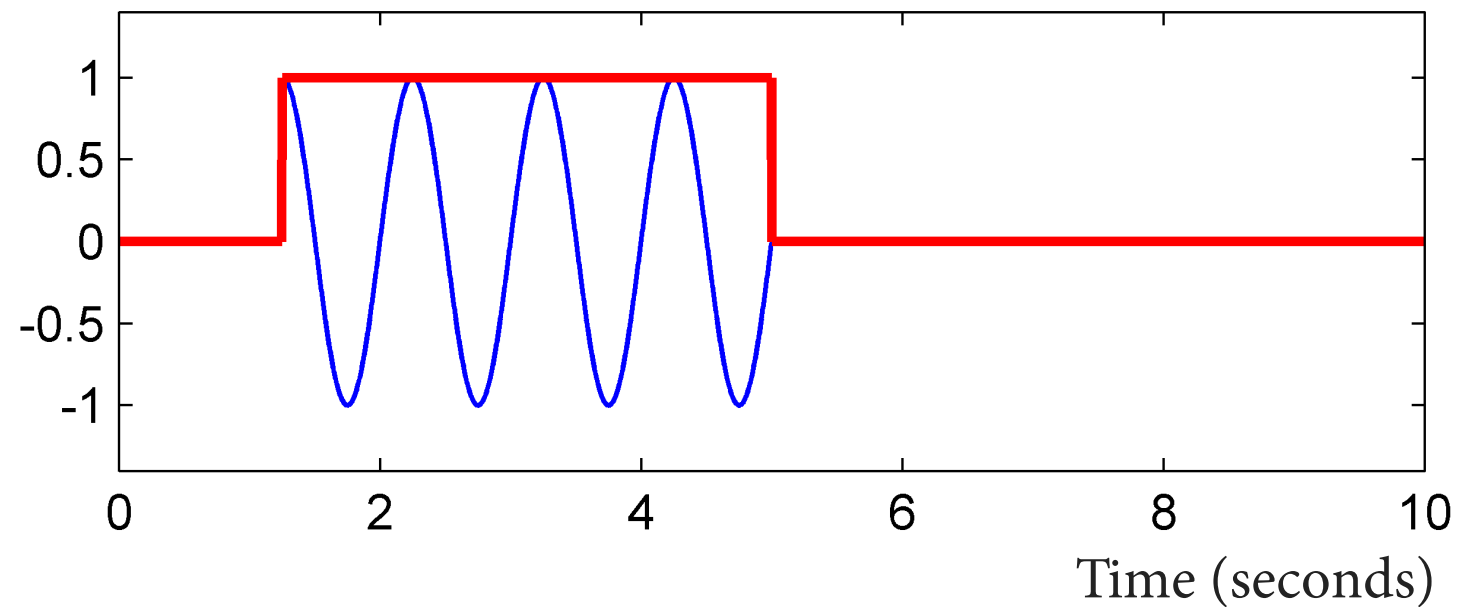
Solution



Context

Short-time Fourier transform

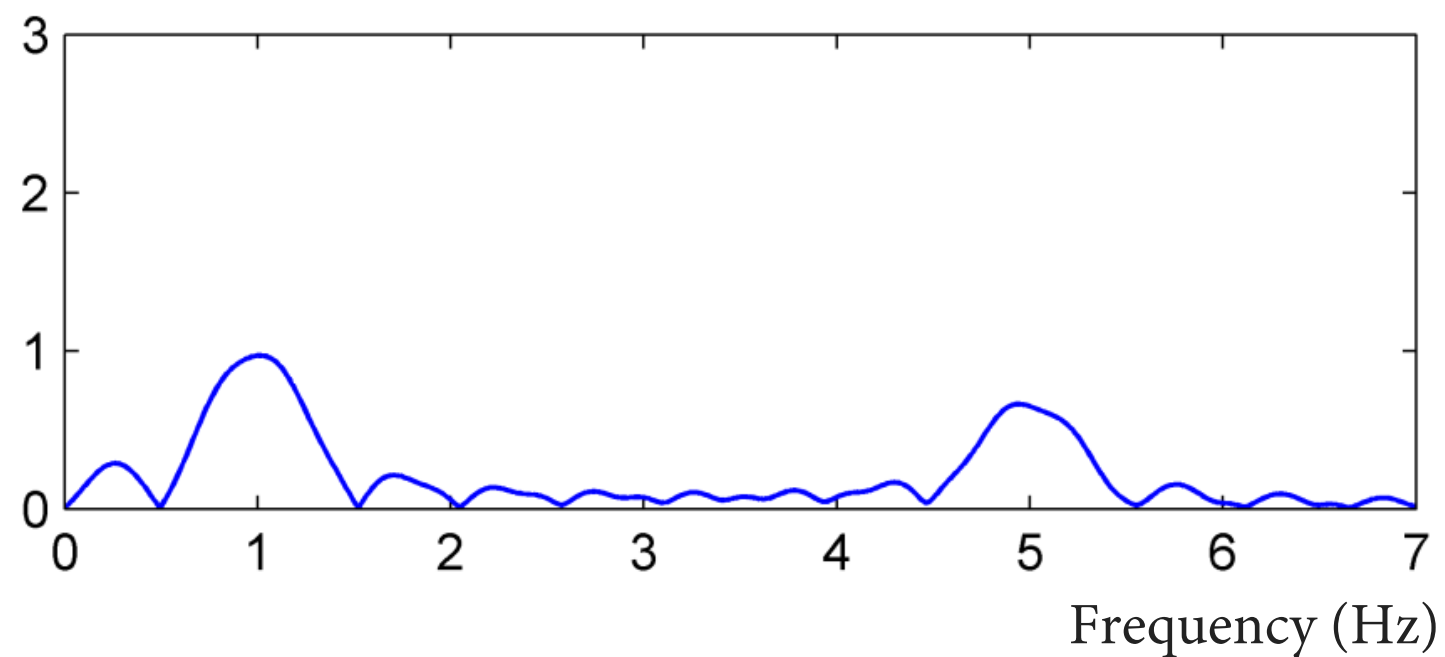
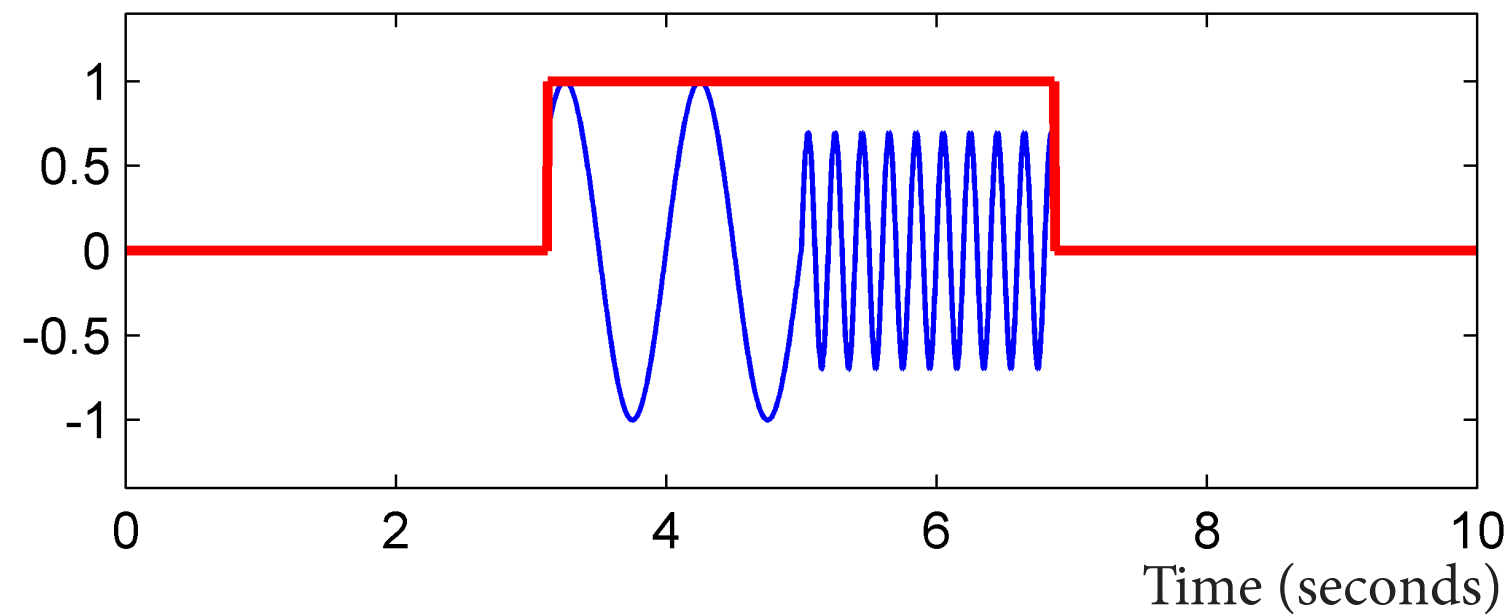
Solution: Consider small segments of the signal and apply FT locally



Context

Short-time Fourier transform

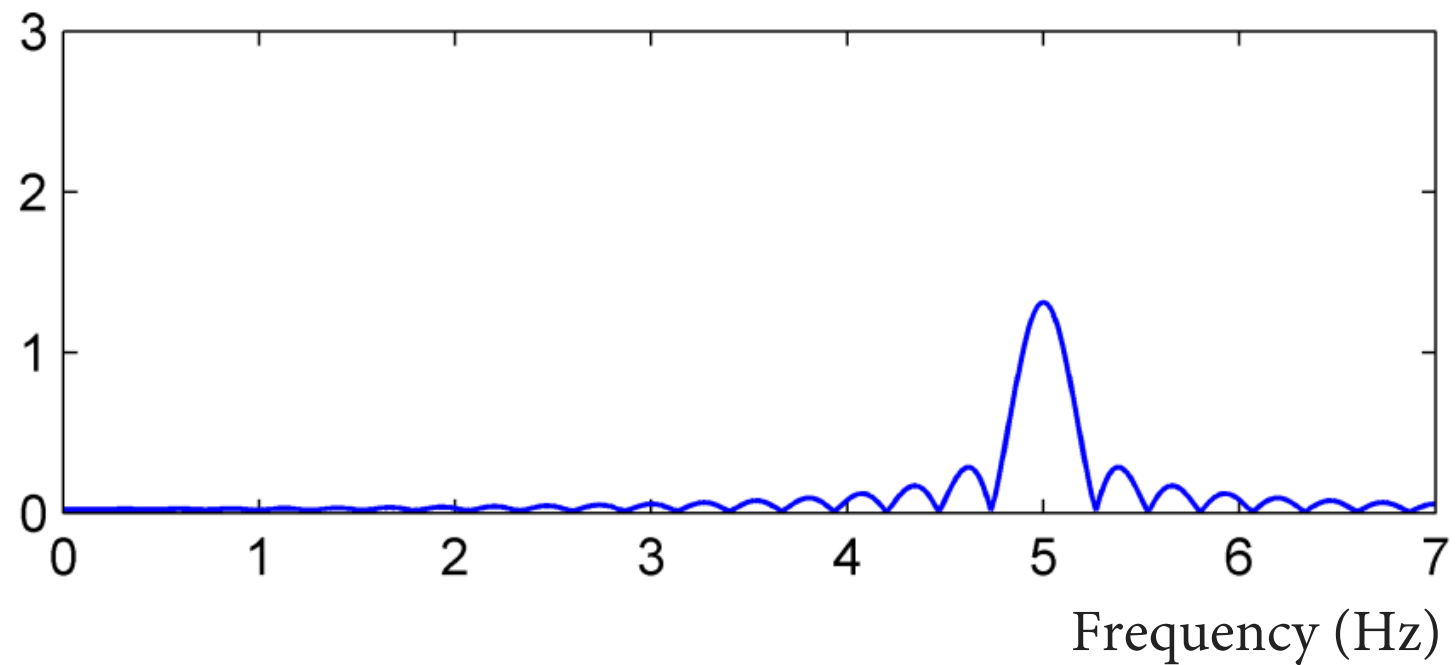
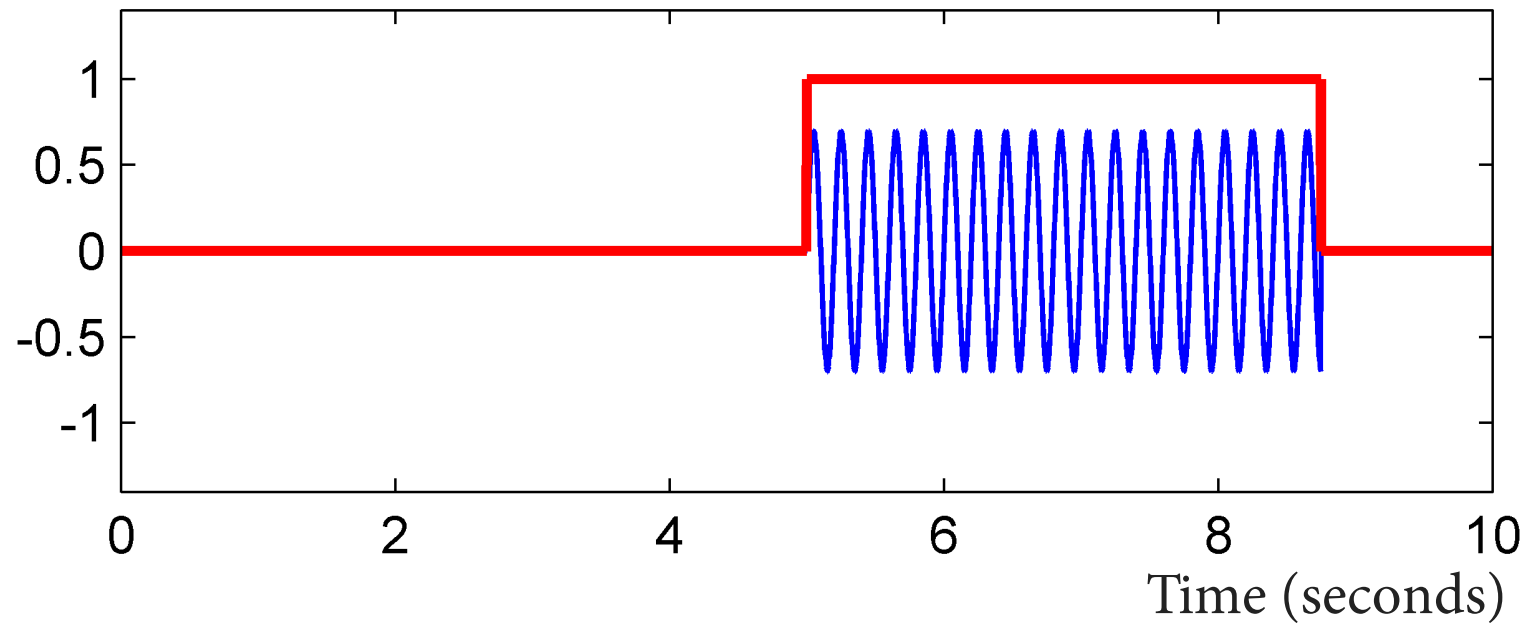
Solution :



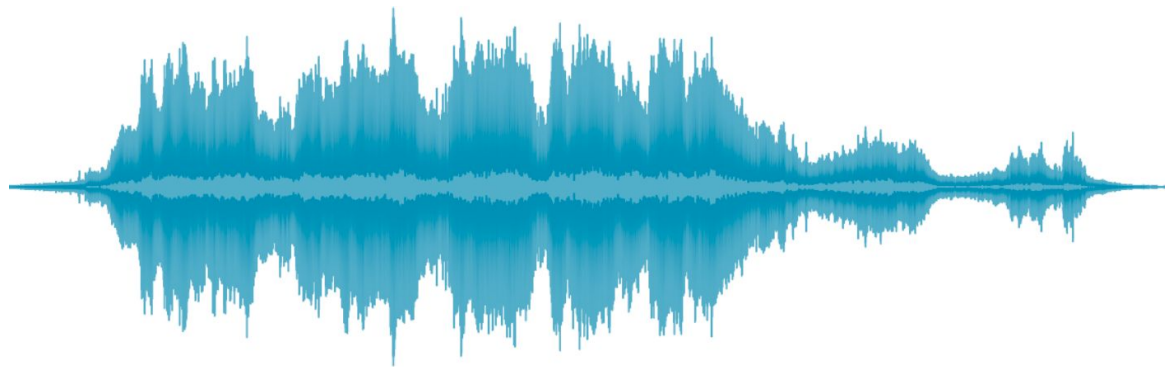
Context

Short-time Fourier transform

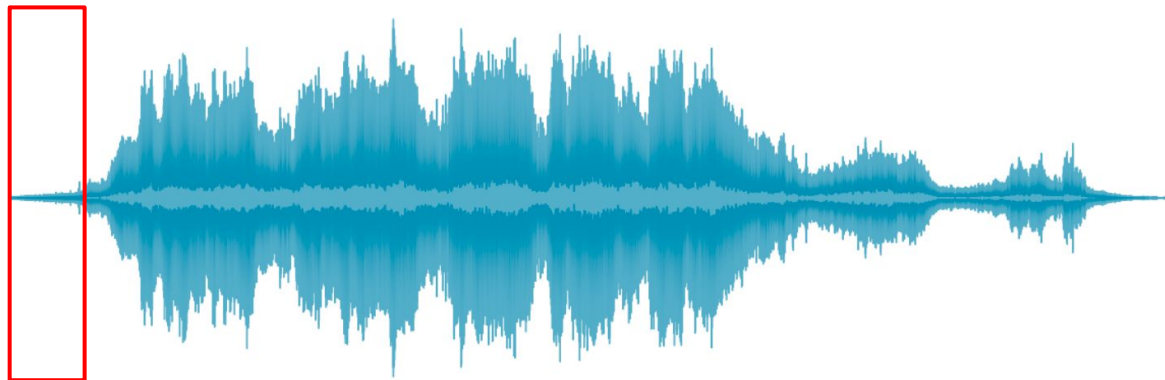
Solution



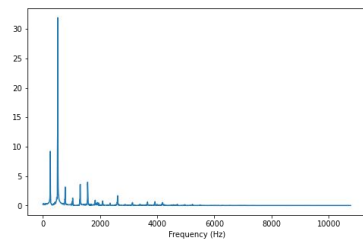
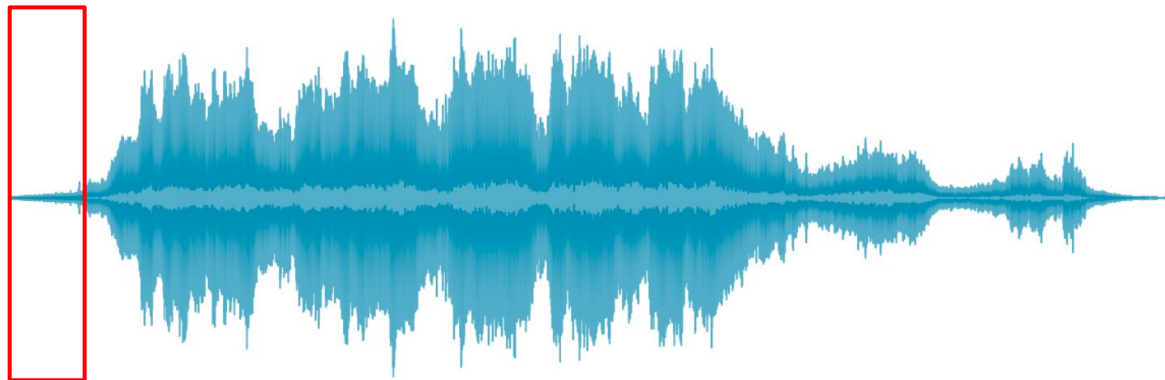
STFT intuition



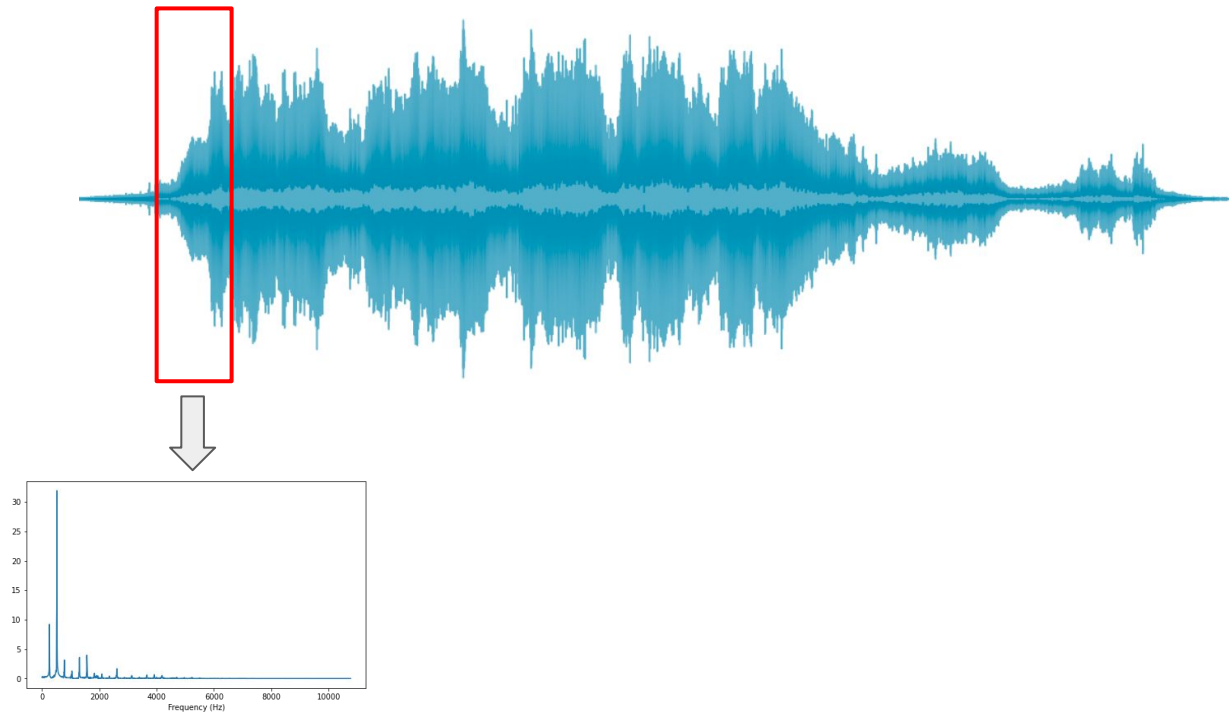
STFT intuition



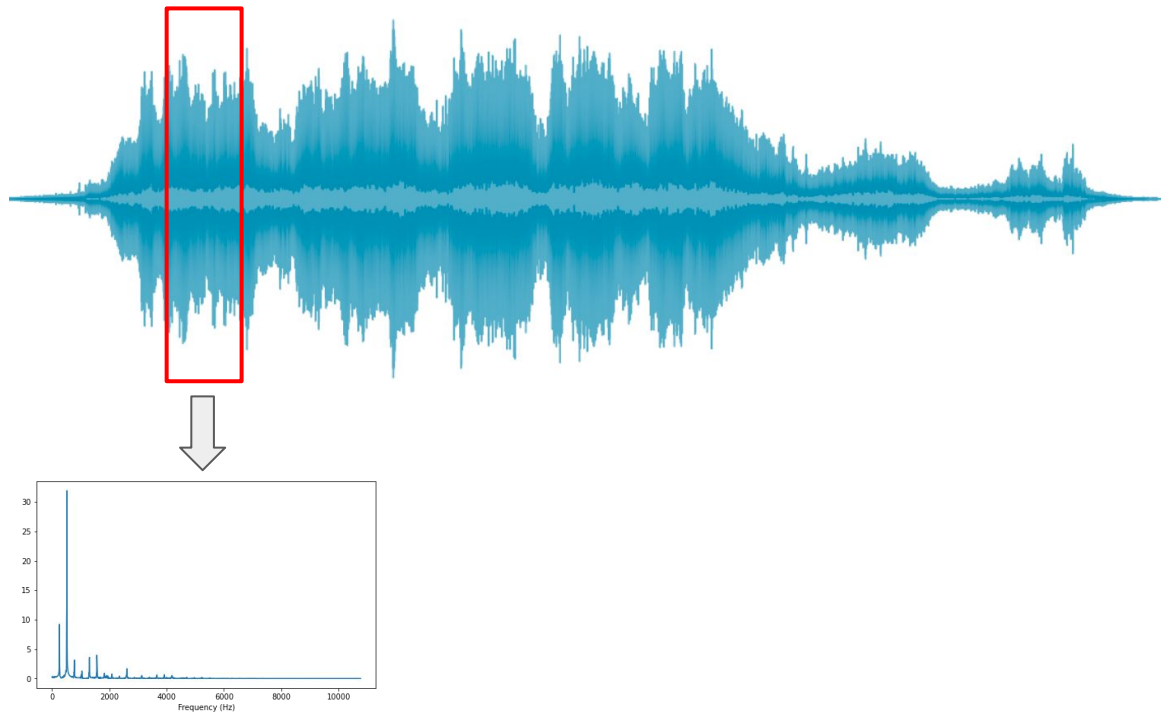
STFT intuition



STFT intuition

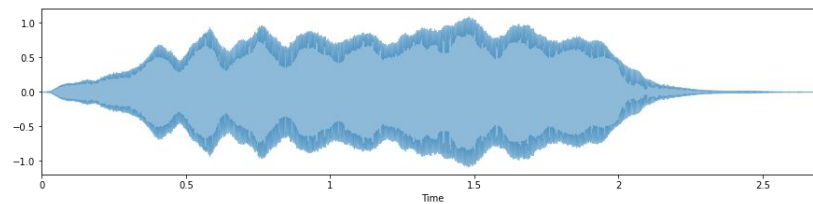


STFT intuition

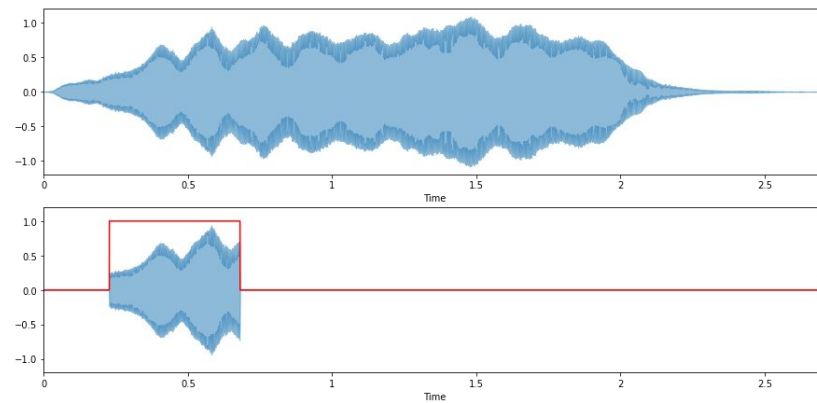


Windowing

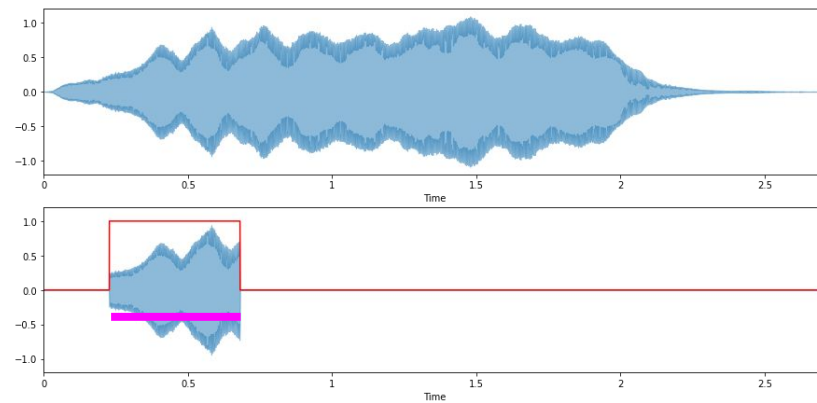
Apply windowing
function to signal



Windowing

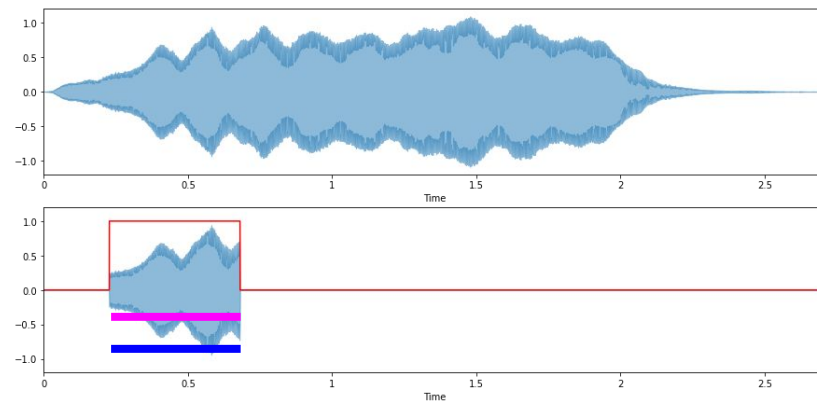


Windowing



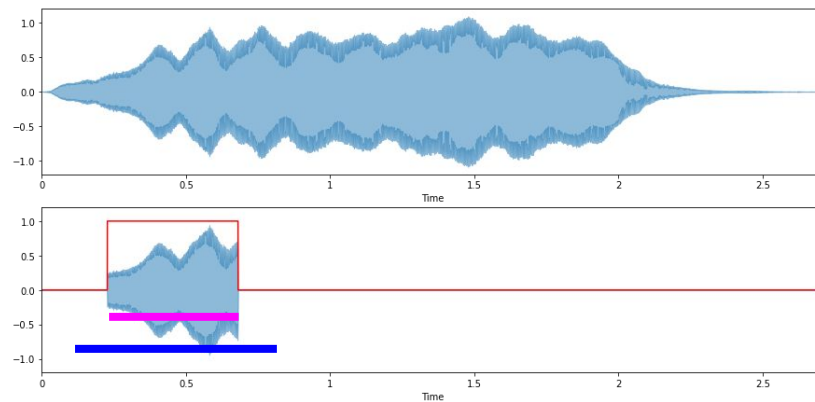
window size

Windowing



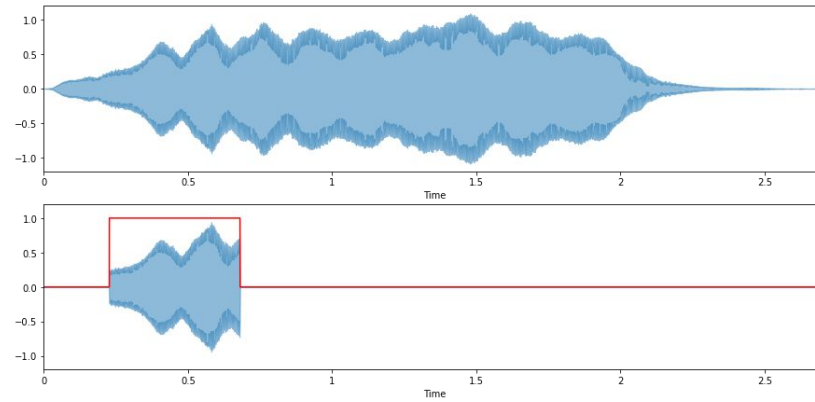
window size = frame size

Windowing

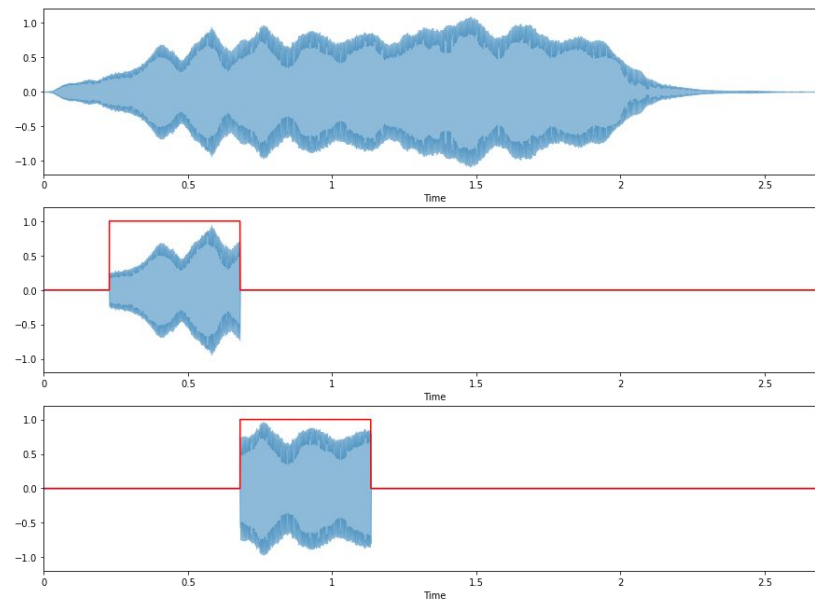


window size $\neq$ frame size (Use of zero padding)

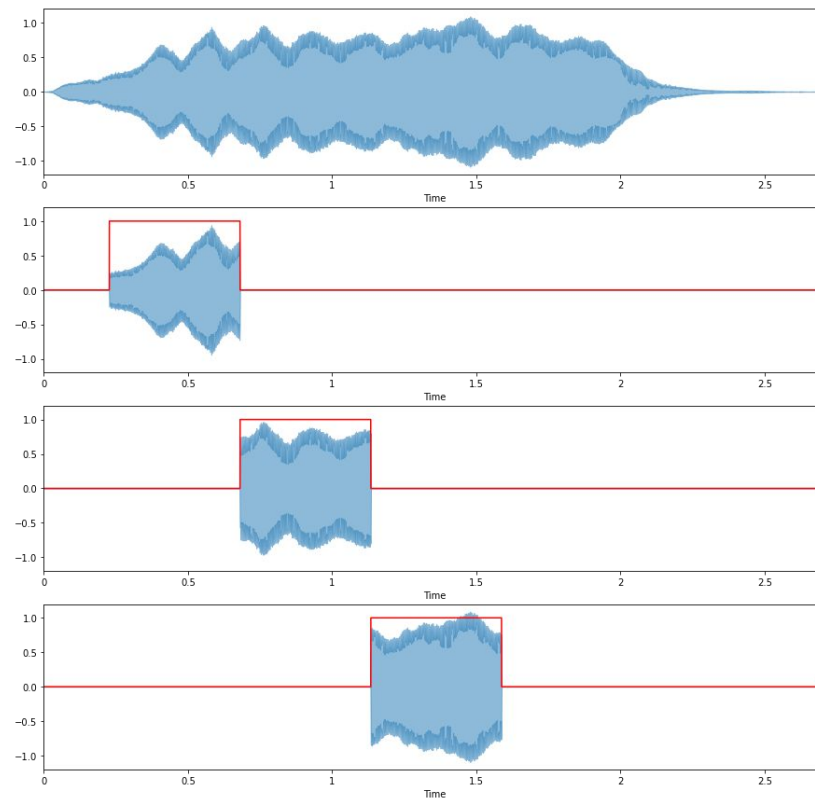
STFT



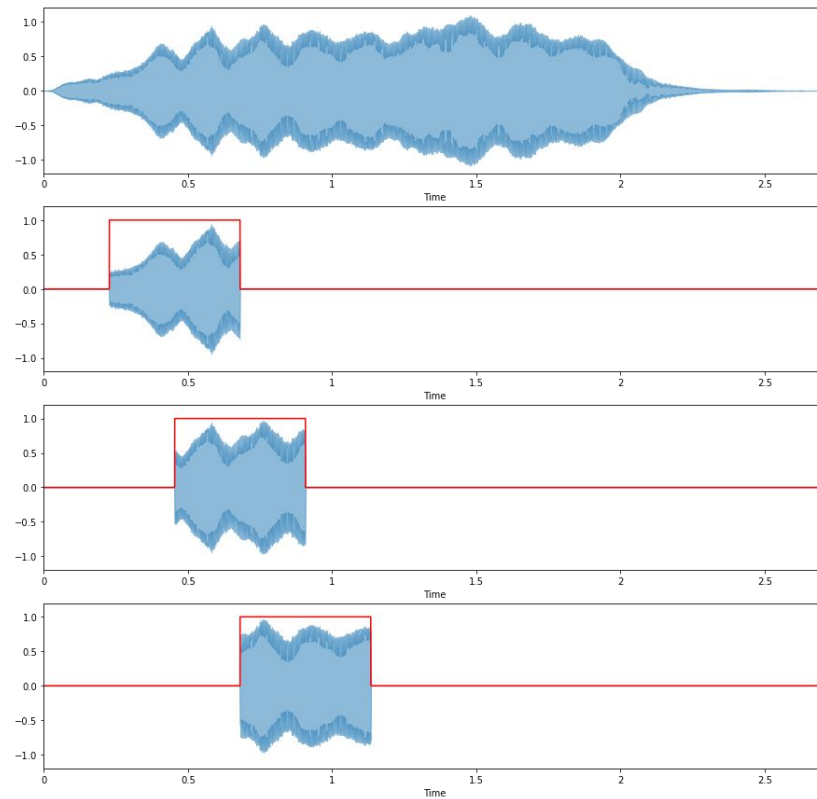
STFT



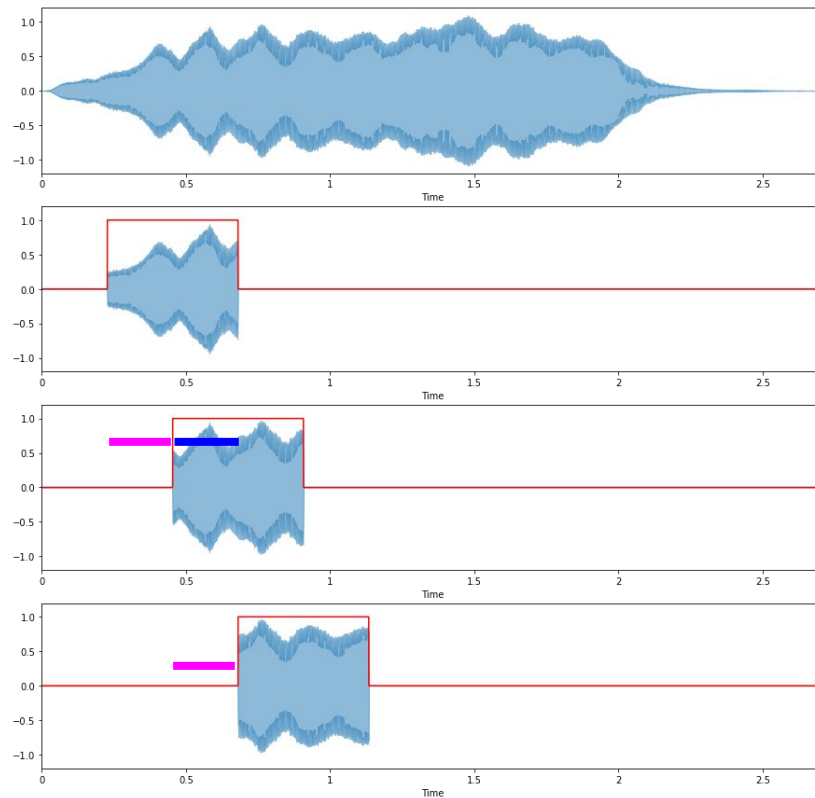
STFT



Overlapping frames



Overlapping frames



hop size (H)

Overlapping size =
window size – hop size

From DFT to STFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}$$

$$S[m, k] = \sum_{n=0}^{N-1} x[n] \cdot w[n - mH] \cdot e^{-j \frac{2\pi kn}{N}}$$

time index (yellow arrow pointing to m)

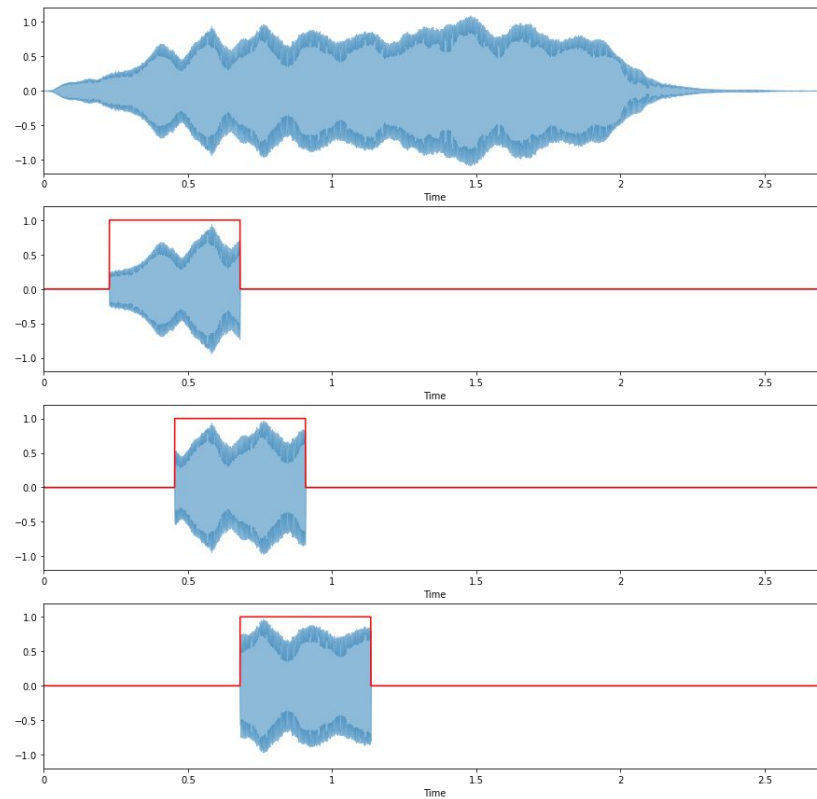
frequency index (red arrow pointing to k)

N points of x (green arrow pointing to n)

window starting at mH (green arrow pointing to $n - mH$)

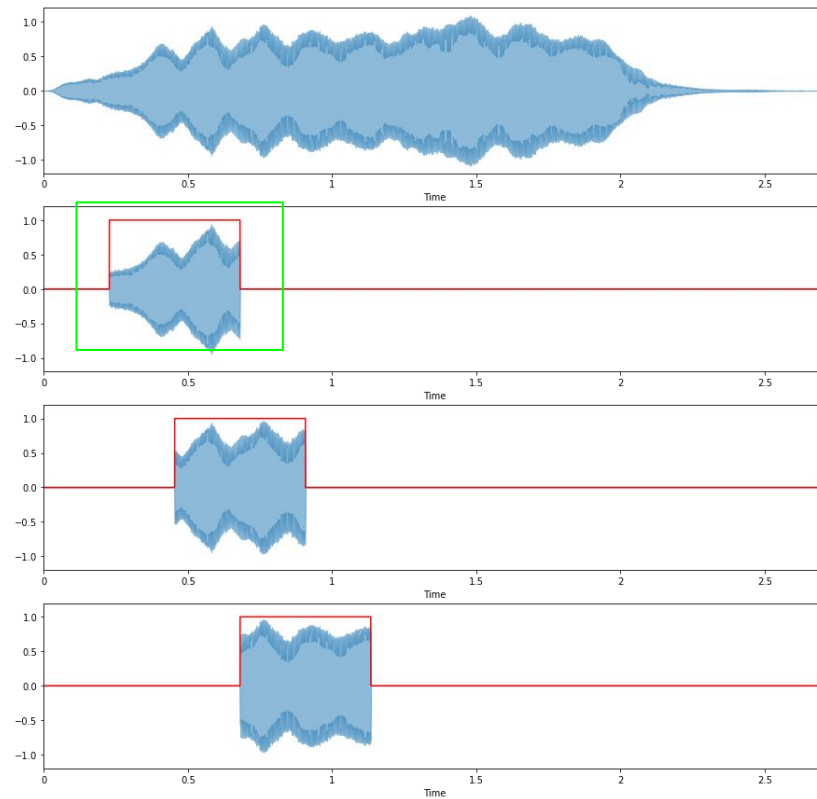
DFT kernel (green arrow pointing to $e^{-j \frac{2\pi kn}{N}}$)

From DFT to STFT

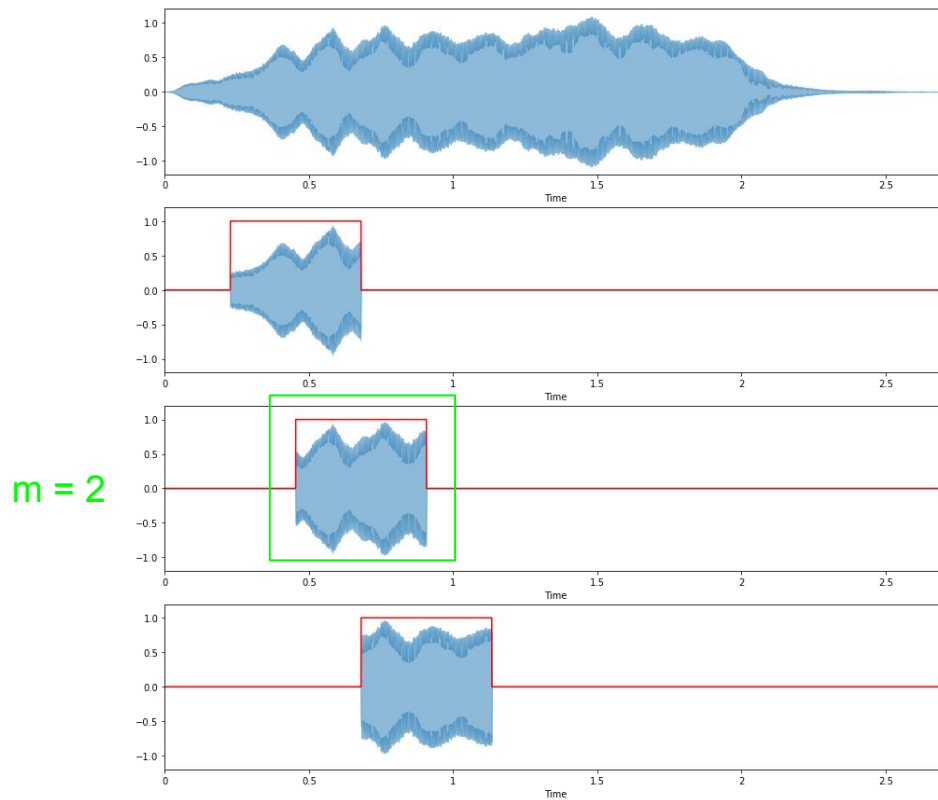


From DFT to STFT

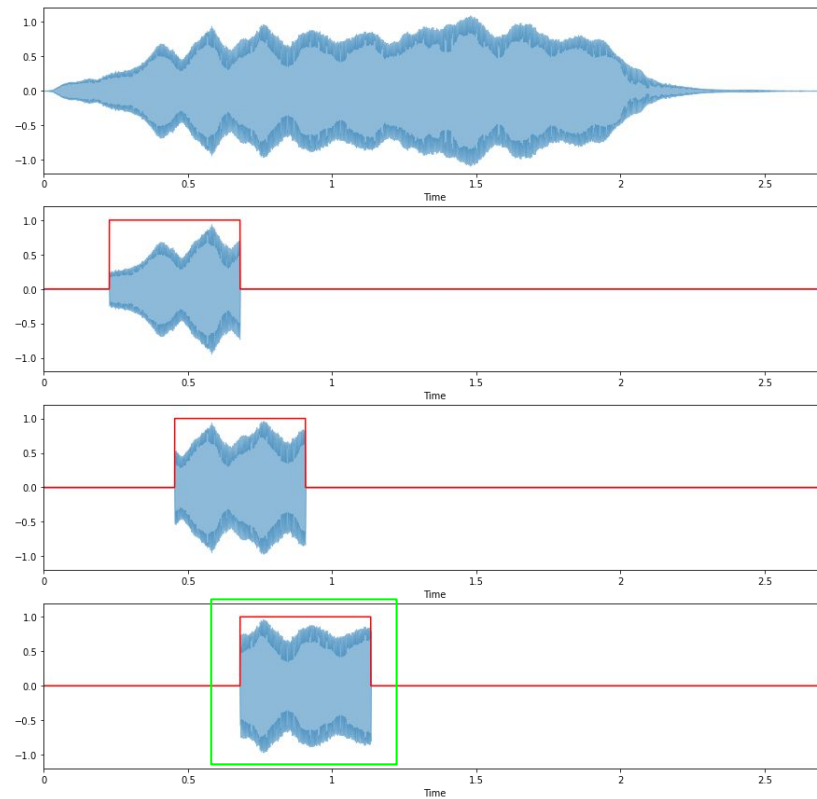
$m = 1$



From DFT to STFT



From DFT to STFT



$m = 3$

Outputs

- DFT
 - Spectral vector (# frequency bins)
 - N complex Fourier coefficients
- STFT
 - Spectral matrix (# frequency bins, # frames)
 - Complex Fourier coefficients

Outputs

$$\# \text{ frequency bins} = \frac{\textit{framesize}}{2} + 1$$

$$\# \text{ frames} = \frac{\textit{samples} - \textit{framesize}}{\textit{hopsize}} + 1$$

Example STFT output

- Signal = 10K samples
- Frame size = 1000
- Hop size = 500

frequency bins = $1000 / 2 + 1 = 501 \rightarrow (0, \text{sampling rate}/2)$

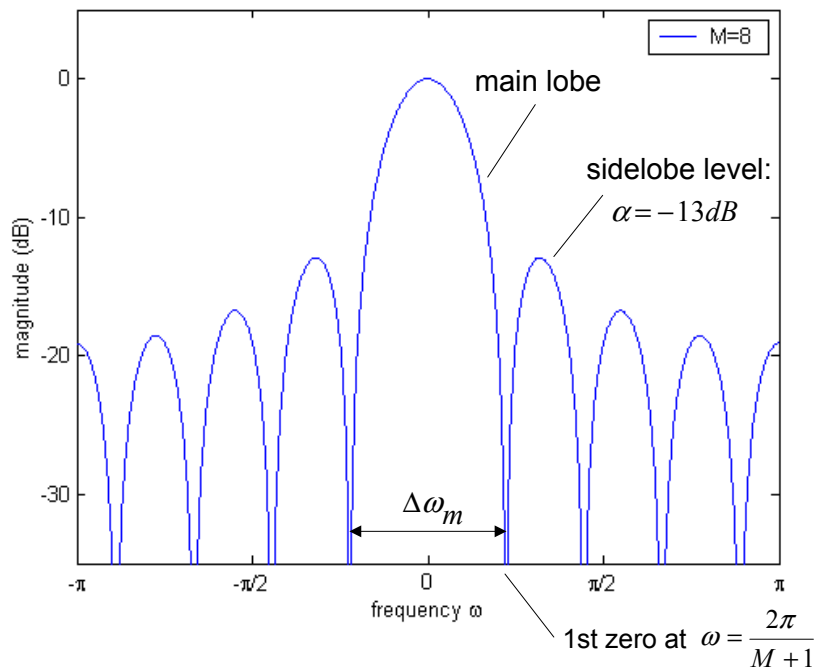
frames = $(10000 - 1000) / 500 + 1 = 19$

STFT parameters

- Frame size 512, 1024, 2048, 4096, 8192
- Hop size 256, 512, 1024, 2048, 4096 $\frac{1}{2}$ K, $\frac{1}{4}$ K, $\frac{1}{8}$ K
- Windowing
function

Windowing

Recall



For a rectangular window $w_R[n] = 1$, $0 \leq n \leq M$,

$$\begin{aligned} W_R(\omega) &= \sum_{n=0}^M e^{-j\omega n} \\ &= e^{-j\omega M/2} \left(\frac{\sin(\omega(M+1)/2)}{\sin(\omega/2)} \right) \end{aligned}$$

The corresponding magnitude spectrum illustrated here for $M = 8$:

$|W_R(\omega)|$ is characterized by a **main lobe**, centered at $\omega = 0$, and secondary lobes of lower amplitudes, also called **sidelobes**. For $w_R[n]$, we have:

- mainlobe width: $\Delta\omega_m = \frac{4\pi}{M+1}$
- peak sidelobe level: $\alpha = -13$ dB (independent of M)

Windowing

Time-Frequency trade-offs

When applied, the DTFT window function has two key features:

- **Main lobe width:** Indicates frequency resolution. Narrower main lobe $\rightarrow$ better ability to distinguish close frequencies.
- **Sidelobe attenuation:** Indicates how much leakage is reduced. Lower sidelobes $\rightarrow$ less spectral leakage.

Time-frequency Trade-Off consists in:

- Narrow main lobe $\rightarrow$ Higher sidelobes (e.g. Rect. window)
Good frequency resolution and Poor leakage suppression
- Wide main lobe $\rightarrow$ Lower sidelobes (e.g. Blackman window)
Poor frequency resolution and Better leakage suppression

This trade-off is **unavoidable** due to the uncertainty principle in time-frequency analysis — you can't simultaneously localize both perfectly

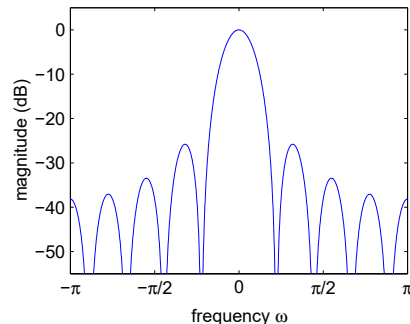
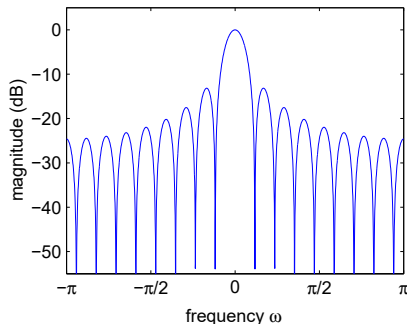
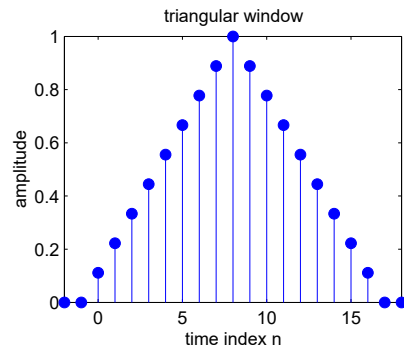
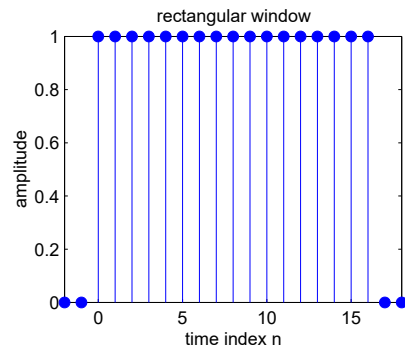


Illustration of the fundamental windowing tradeoff.

Windowing

Comparison for the same window length

Bartlett or triangular window:

$$w[n] = (1 - |\frac{2n}{M} - 1|) w_R[n]$$

Hann Window:

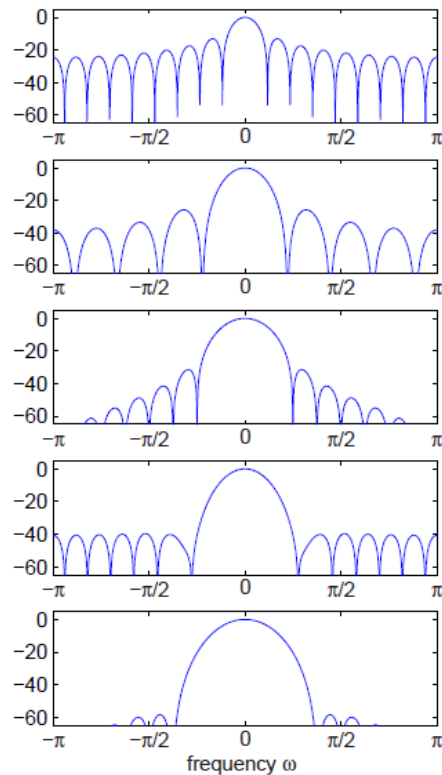
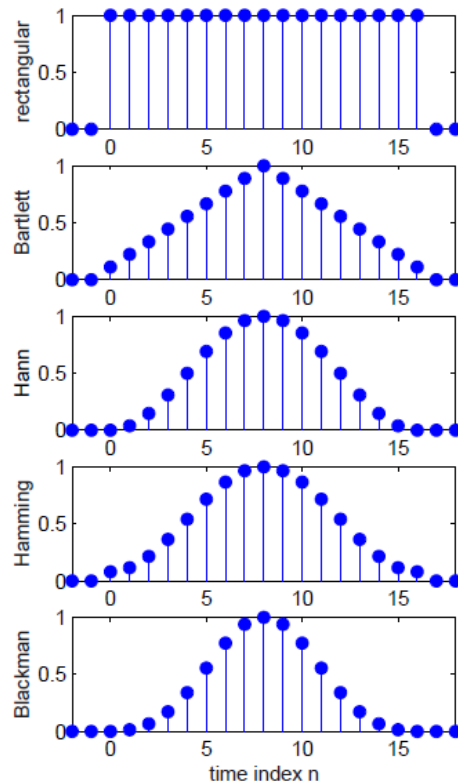
$$w[n] = \frac{1}{2} (1 - \cos \frac{2\pi n}{M}) w_R[n]$$

Hamming Window:

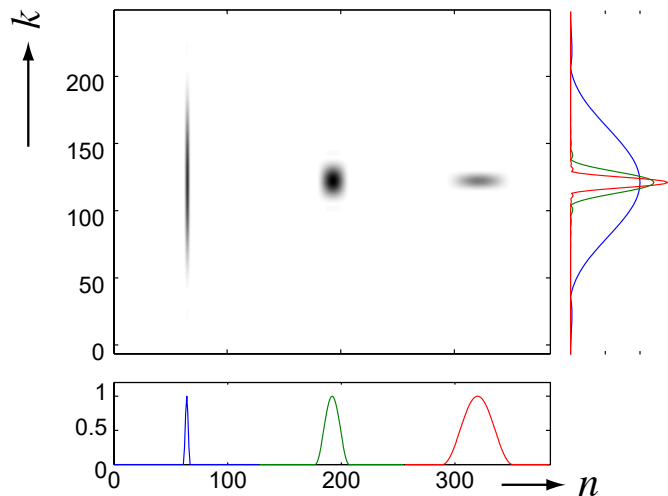
$$w[n] = (0.54 - 0.46 \cos \frac{2\pi n}{M}) w_R[n]$$

Blackman Window:

$$w[n] = (0.42 - 0.5 \cos \frac{2\pi n}{M} + 0.08 \cos \frac{4\pi n}{M}) w_R[n]$$

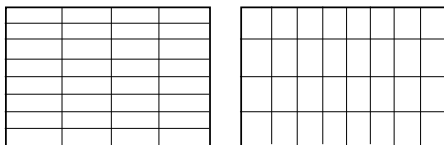
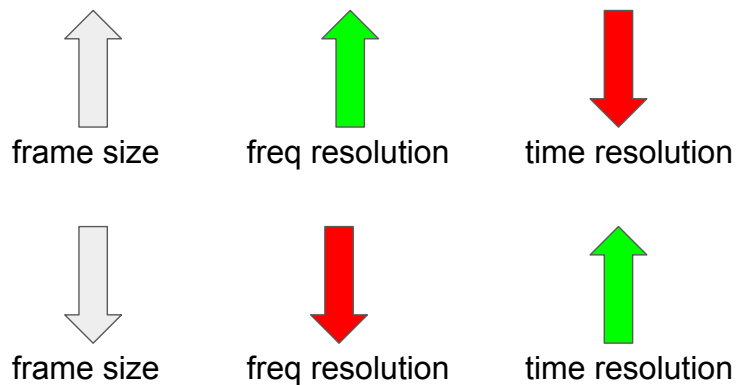


Time / frequency trade off



- Disks show "blurring" due to window length;
- **Area** of disk is constant

Uncertainty principle: $\Delta t \Delta f \geq 1/4\pi$



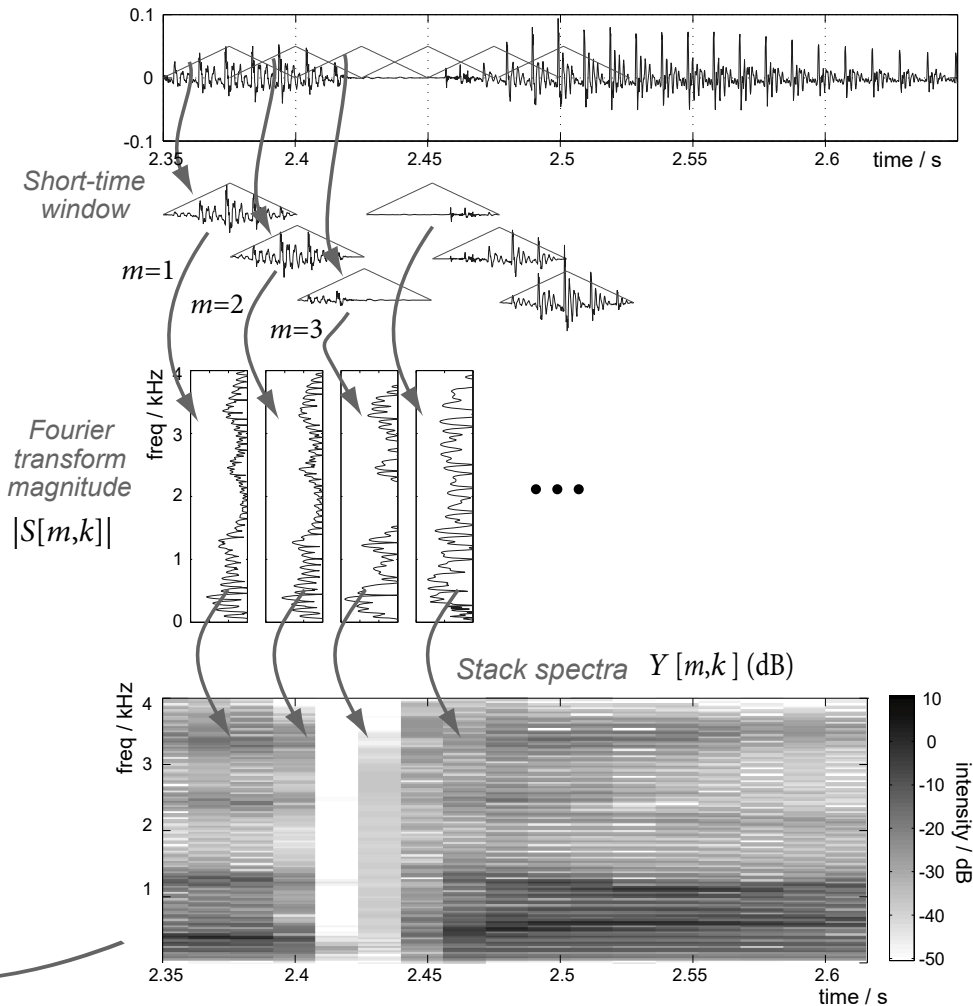
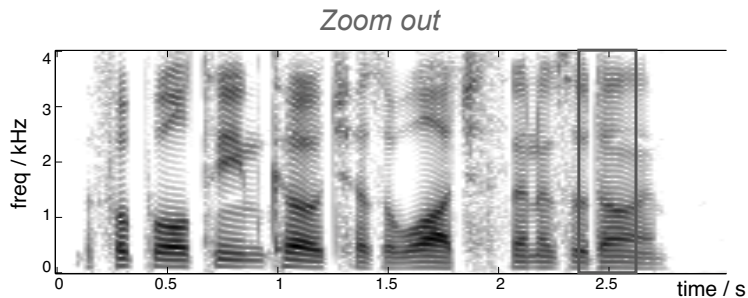
Visualising sound: Spectrogram

Spectrogram (Y)

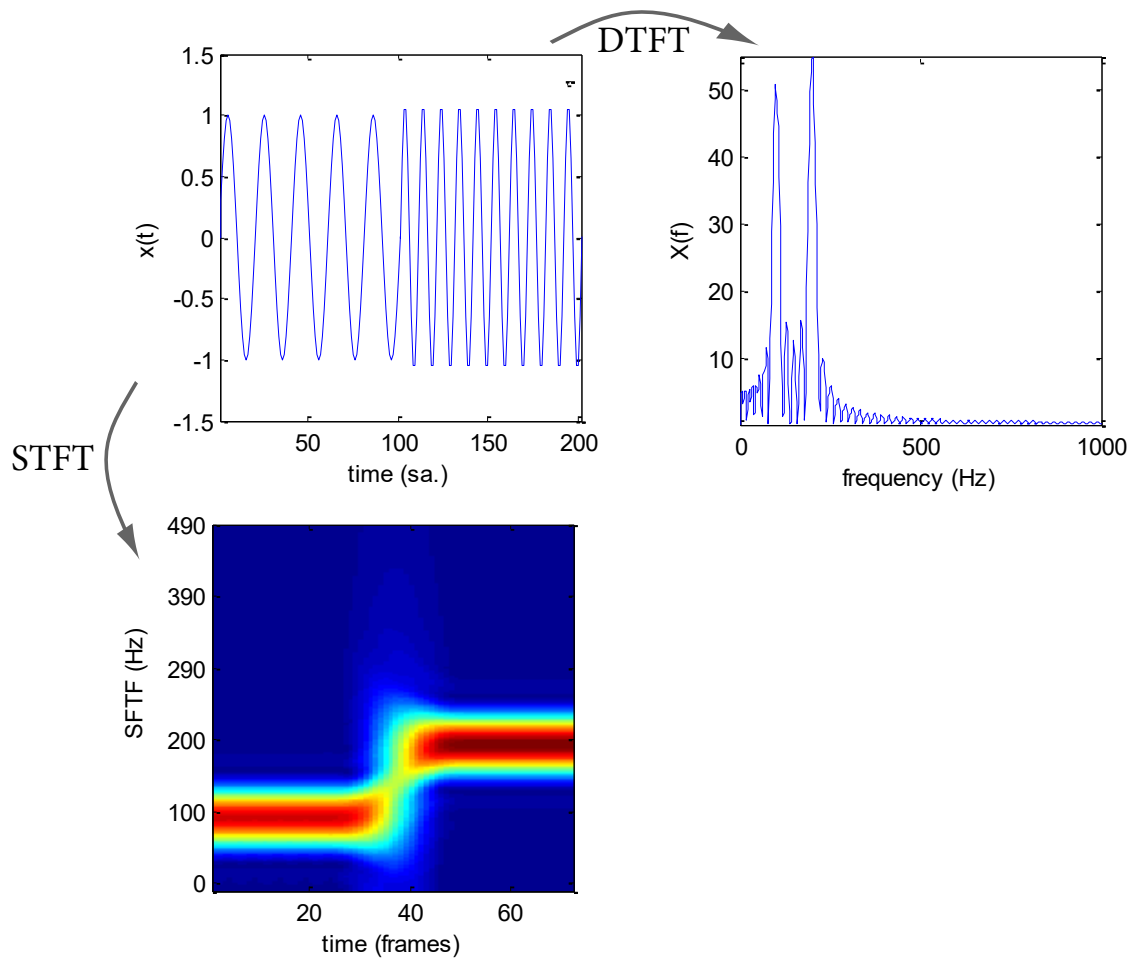
$$Y[m,k] = |S[m,k]|^2$$

$$Y[m,k] \text{ (dB)} = 10 \log_{10} Y[m,k]$$

$$Y[m,k] \text{ (dB)} = 20 \log_{10} |S[m,k]|$$



Visualising sound: Spectrogram

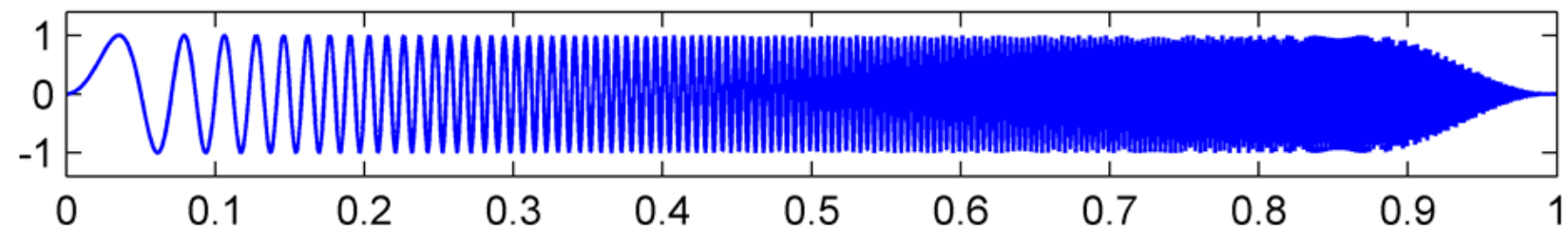


Spectrogram

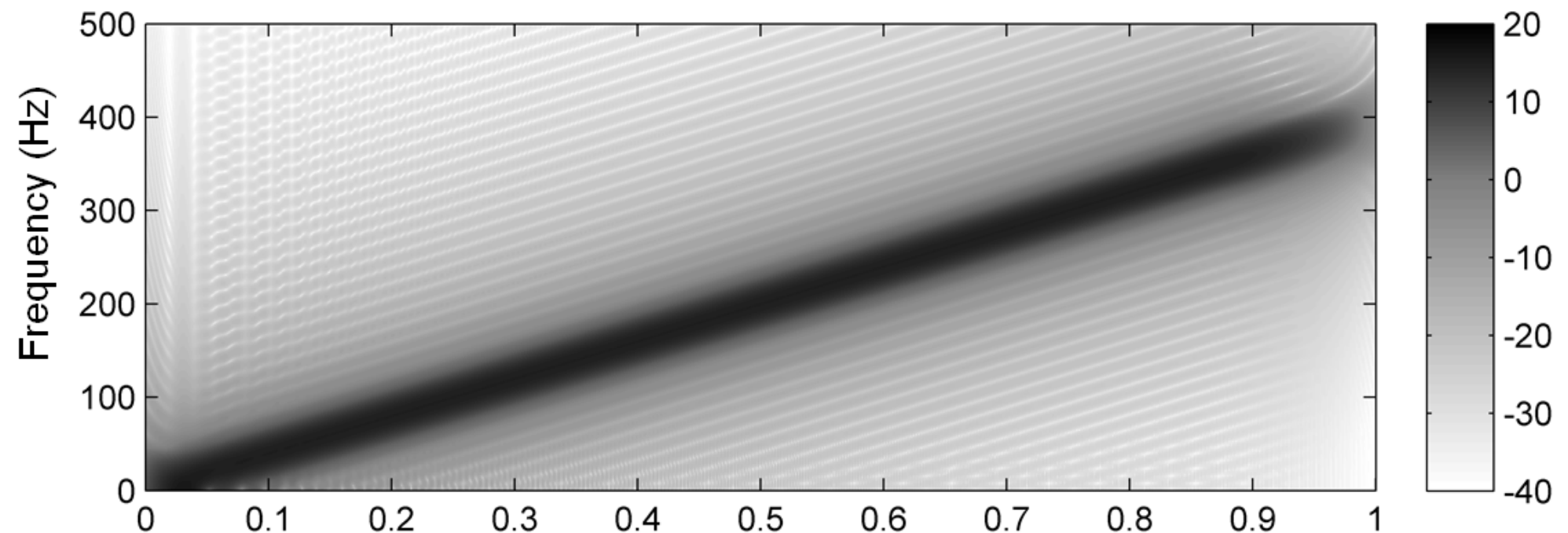
Definition

Example : chirp

Chirp



Sound wave of constant amplitude with linear increase in frequency



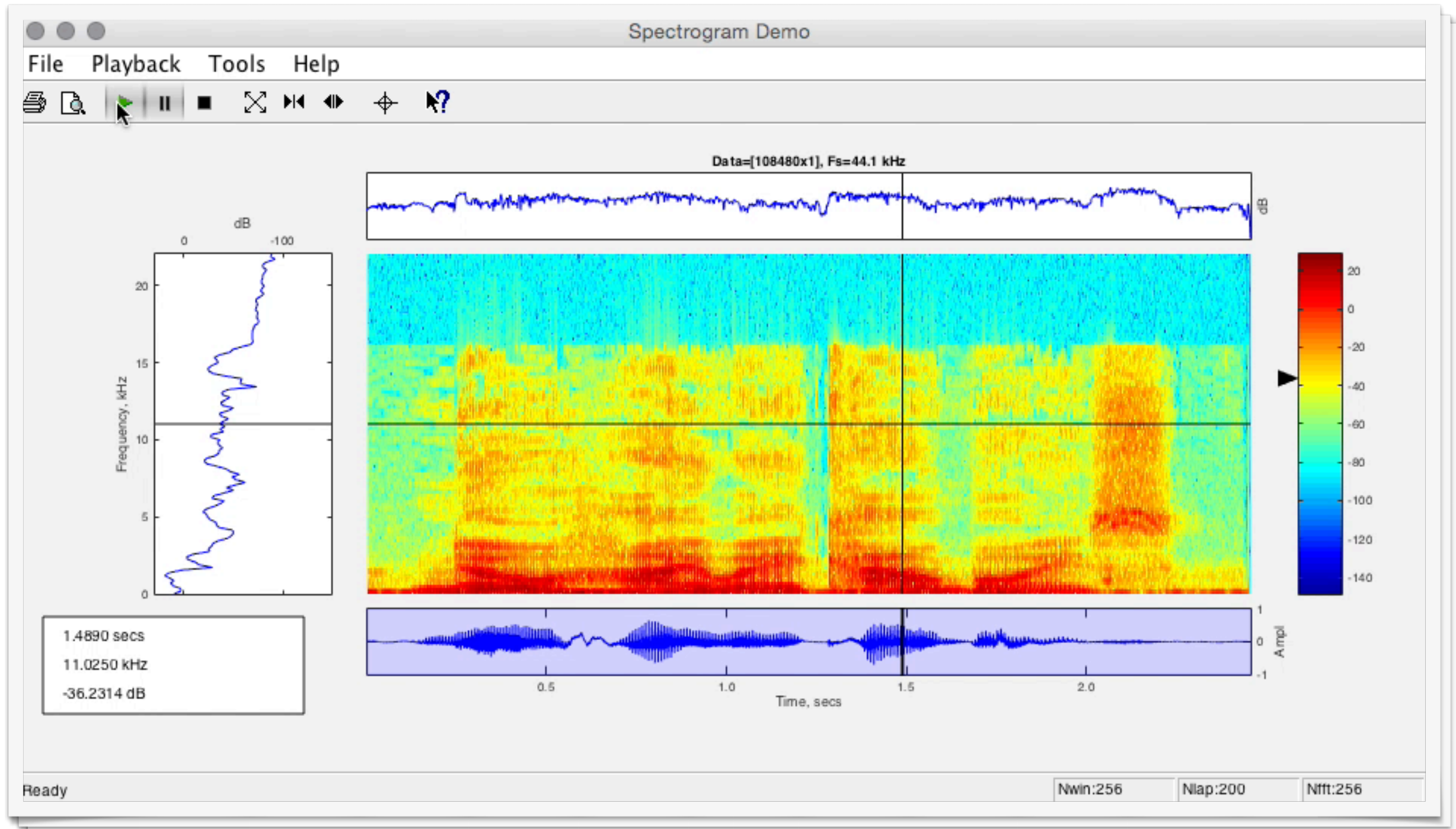
Spectrogram : Hann window

Spectrogram

Definition

Speech signal example

American pronunciation of " No highway cowboys "

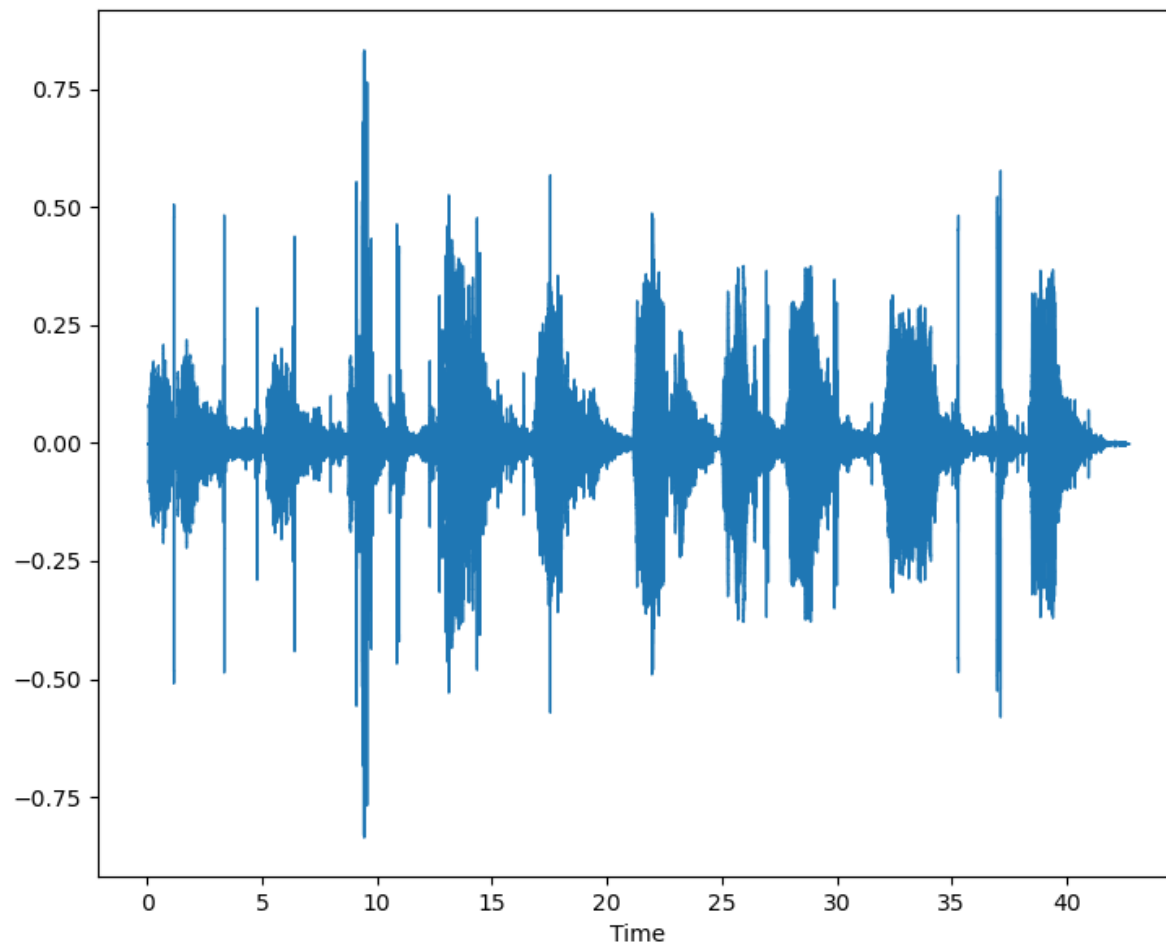


Spectrogram

Definition

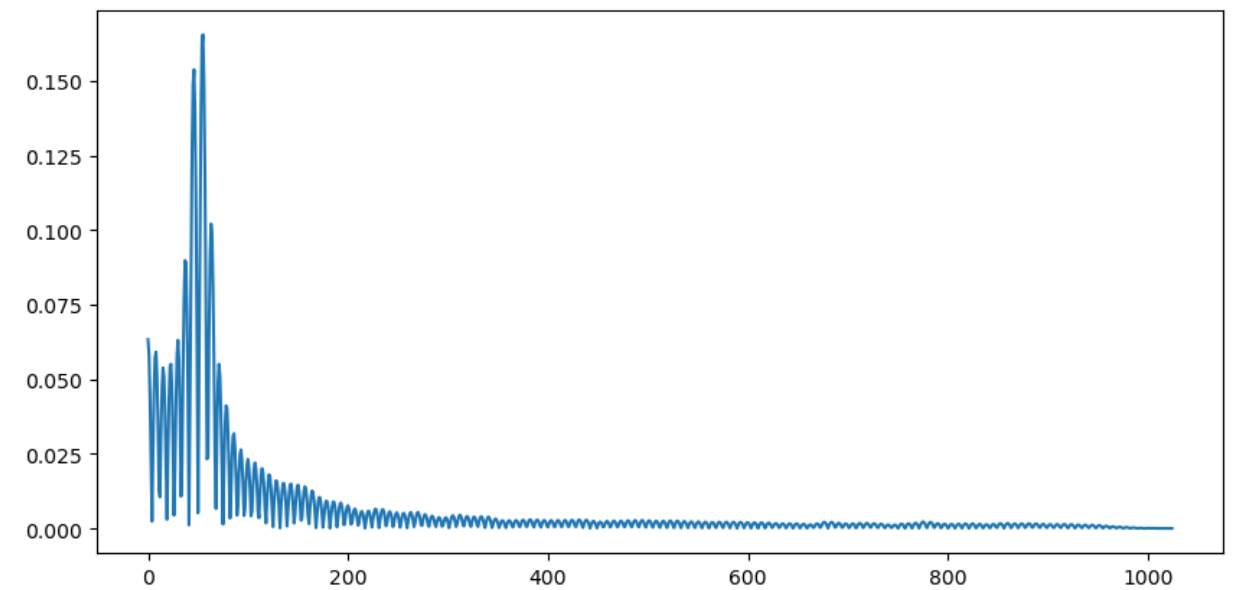
Example : Sealions

Sound wave



Sealions

Overall signal spectrum

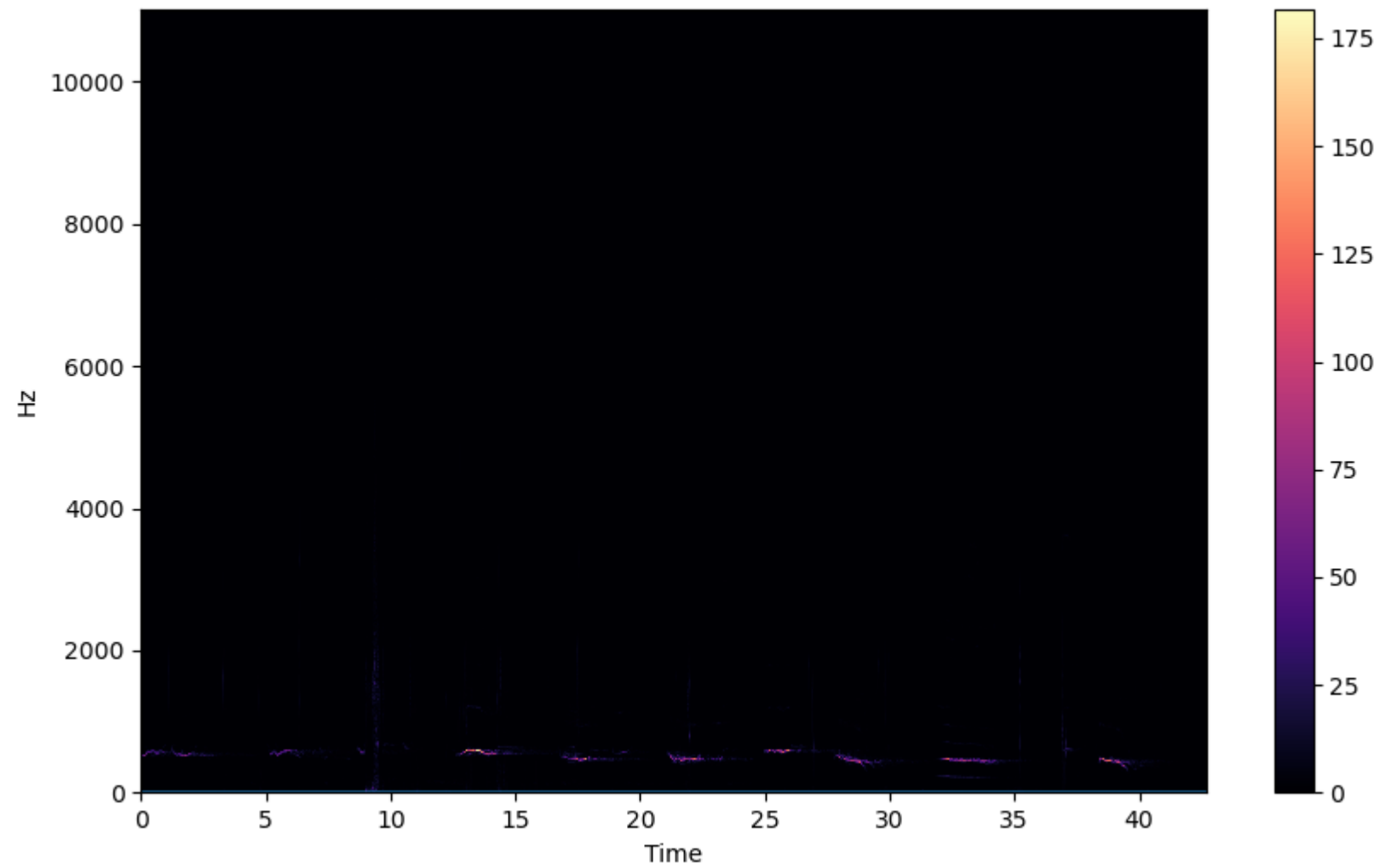


Spectrogram

Definition

Exemple : Sealions

Spectrogram, magnitude

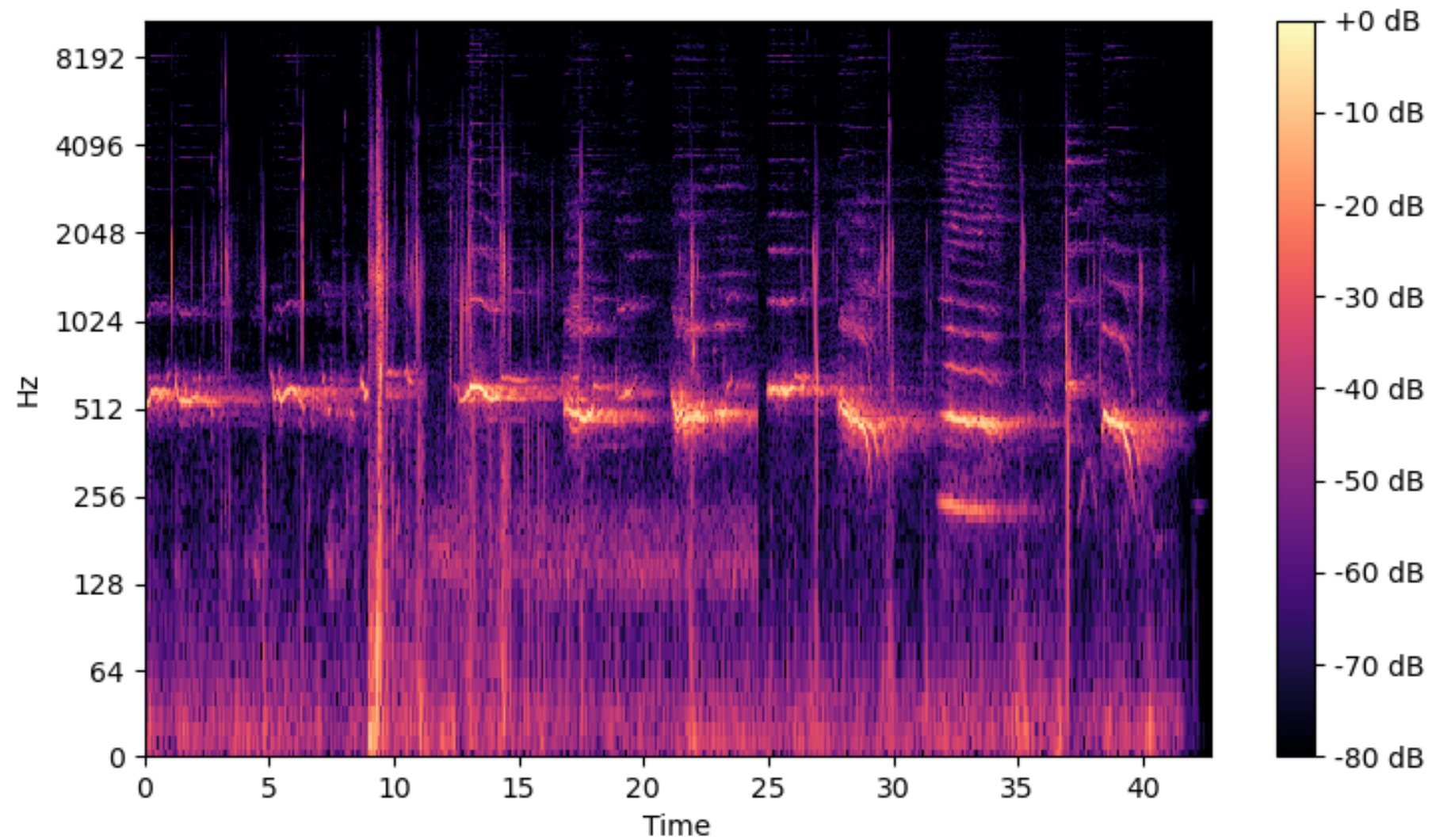


Spectrogram

Definition

Exemple : Sealions

Spectrogram, magnitude in decibels



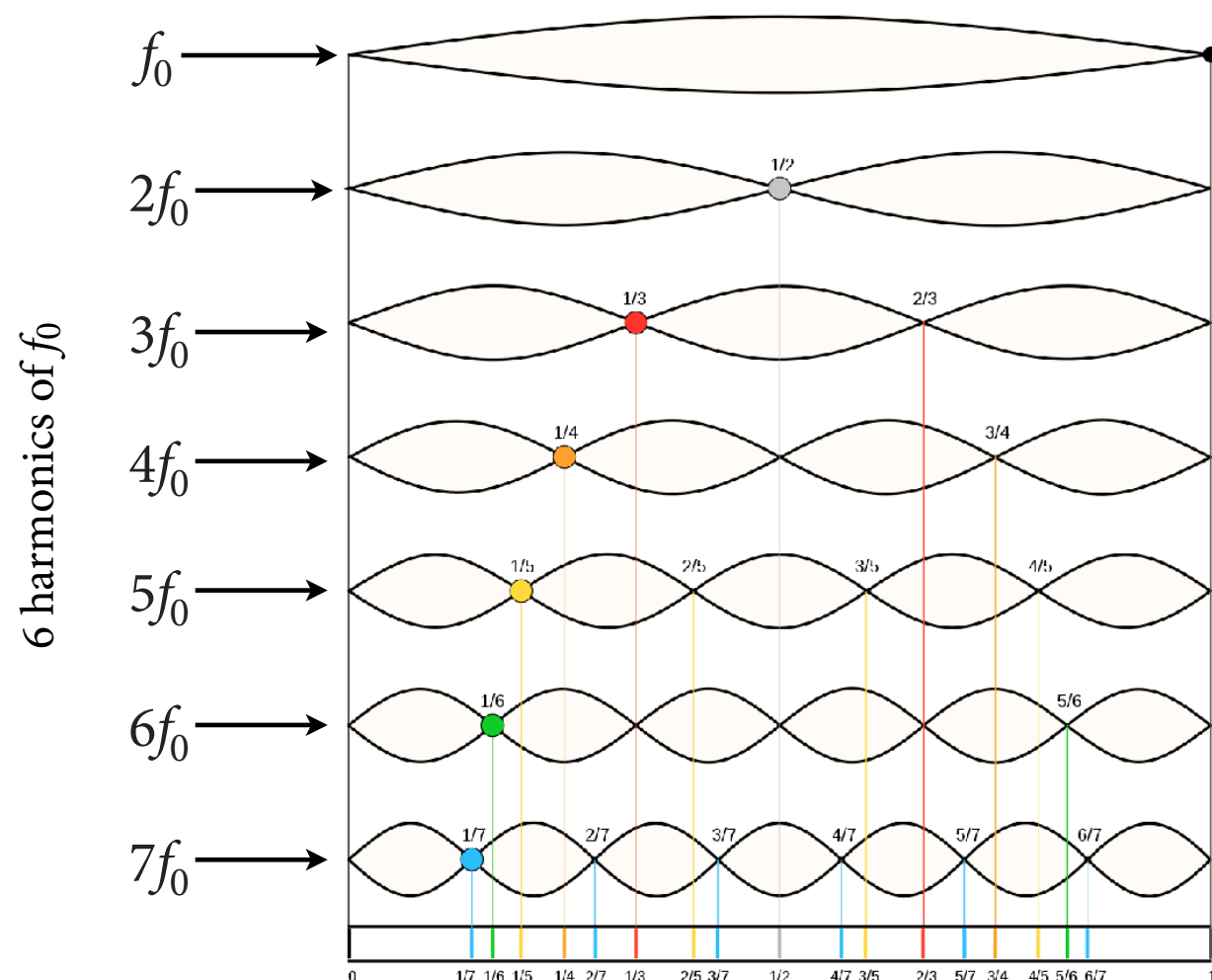
The Role of Window Length in Spectrogram

In time-frequency analysis of audio signals using spectrograms, the window length is a crucial parameter. It determines the trade-off between time resolution and frequency resolution, which significantly affects how features in speech or sound are visualized and interpreted.

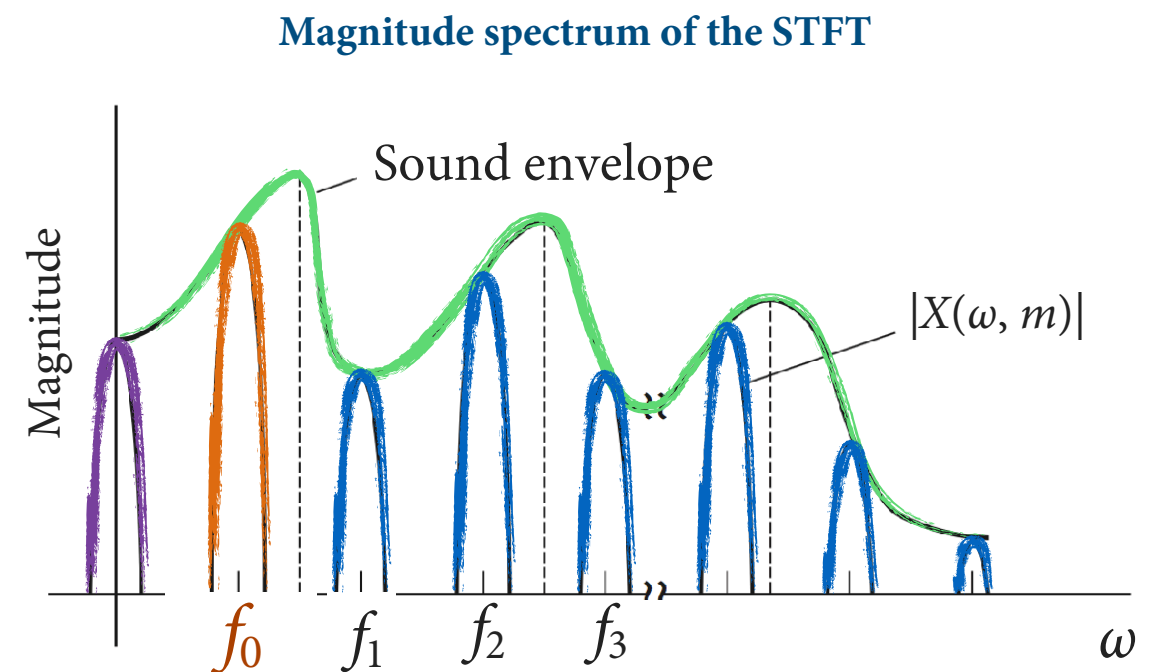
Recall : Fundamental frequency

f_0 : fondamental frequency.

$f_n = n.f_0$: n th order harmonic



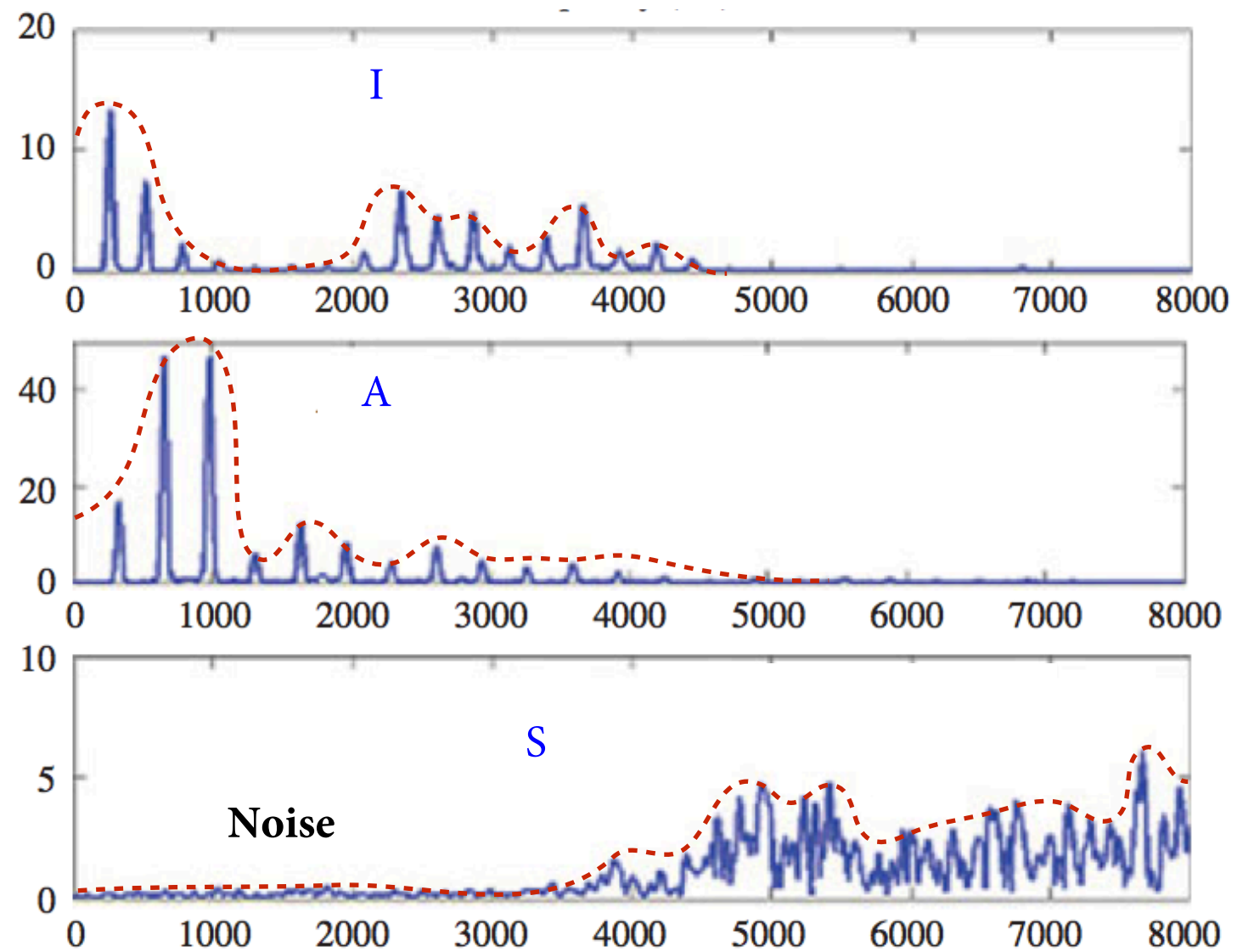
Noise-free sound: Intonation, voice, melodic instrument, etc.



Spectrogram

Fondamental Frequency

Pronunciation of three letters



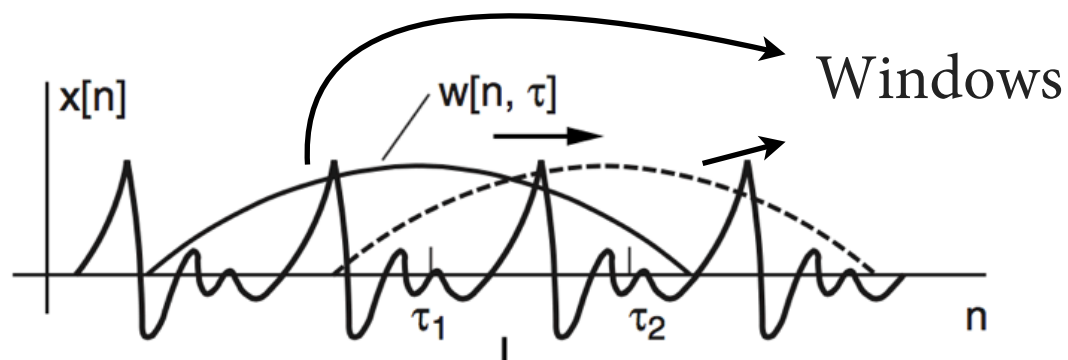
Spectrogram

Remark

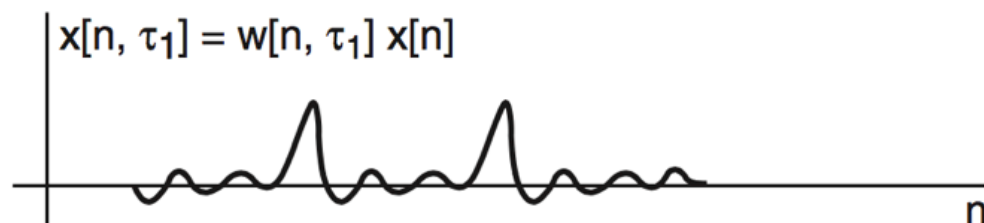
The condition to determine the good window length is due to Nyquist (f_s)

- Wide window $\Leftrightarrow$ Narrowband

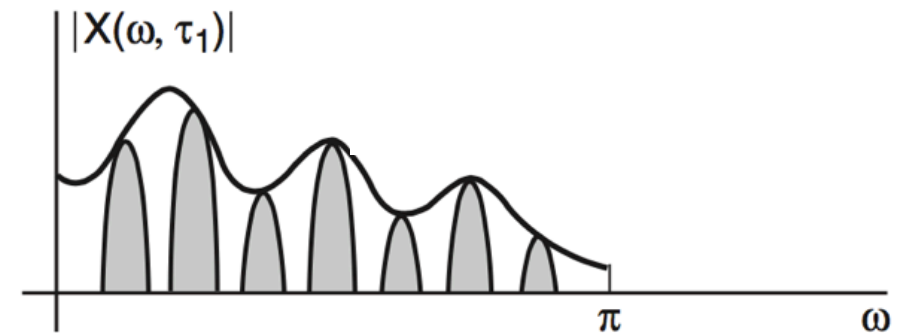
Original signal



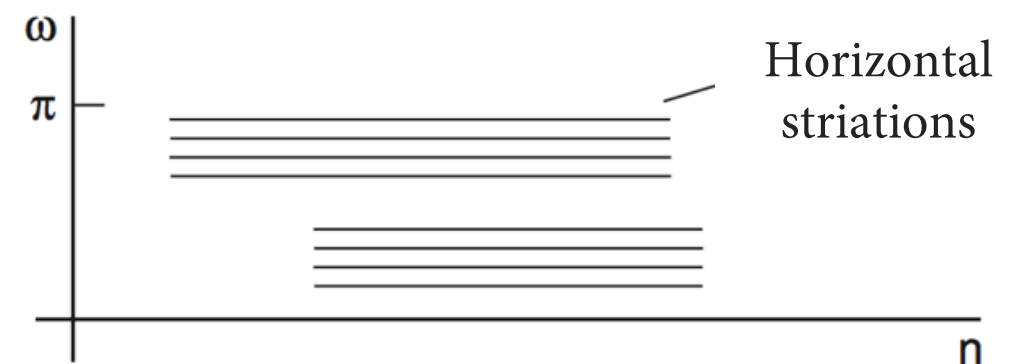
Window signal



STFT



Spectrogram



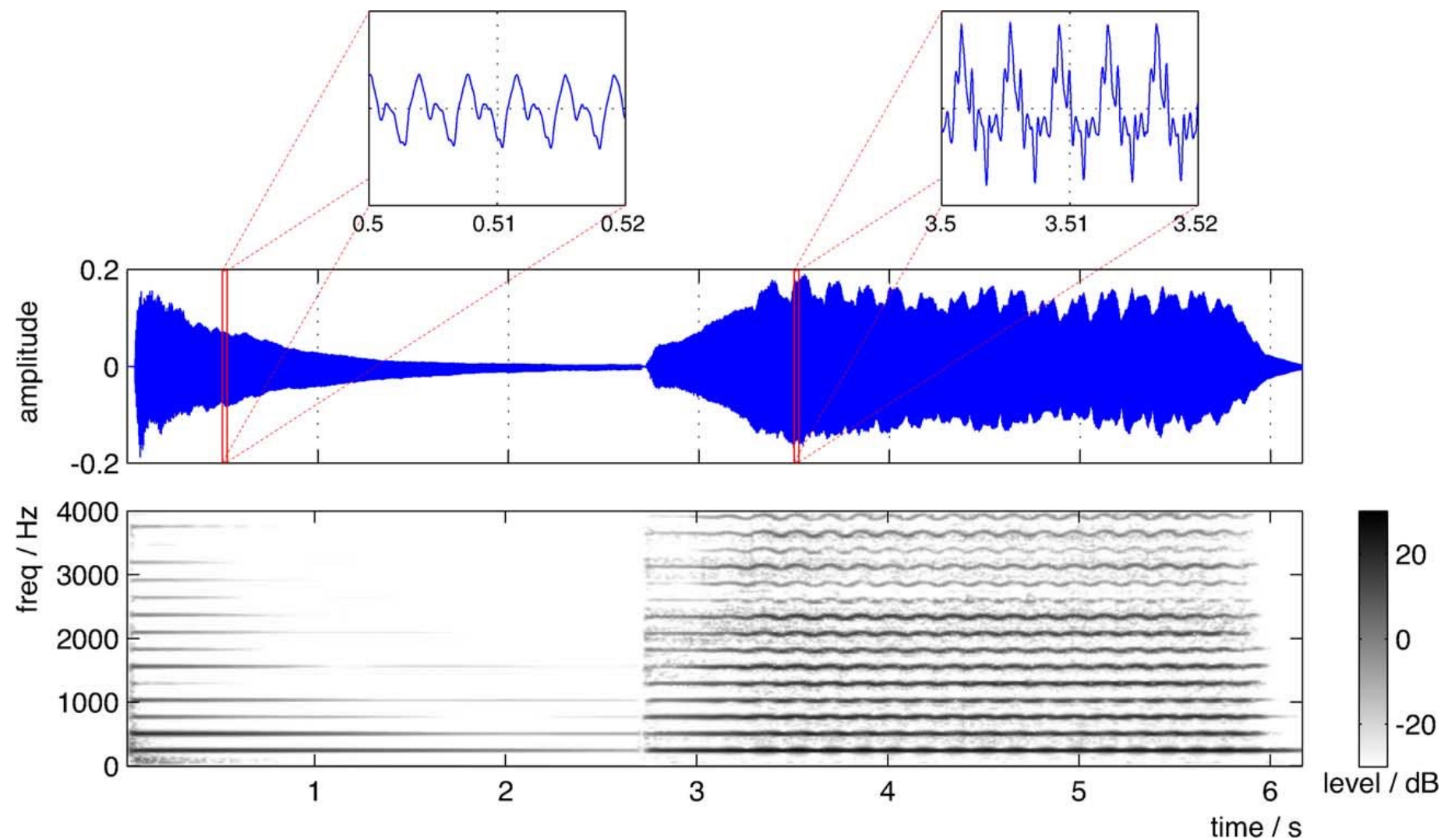
The horizontal streaks in the narrowband spectrogram are multiples of f_0

Spectrogram

Example : music note C

Wide window → Clear harmonics

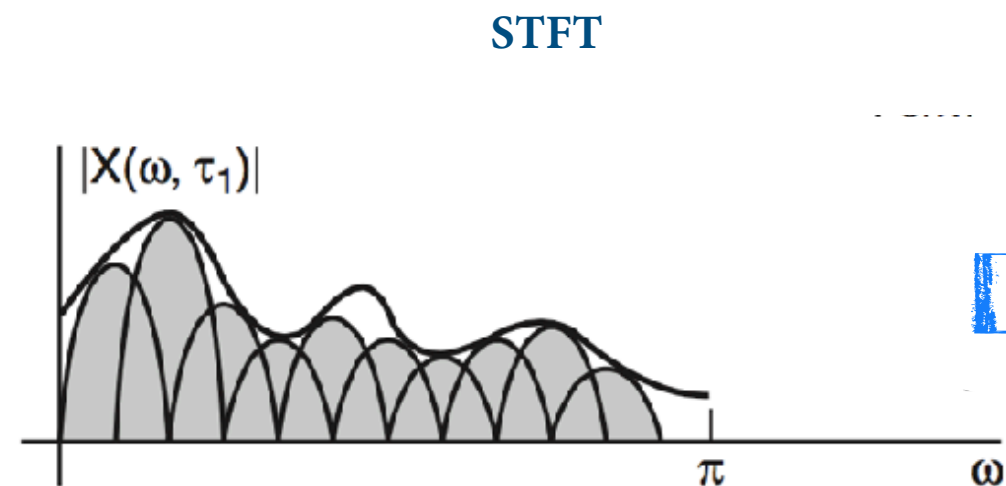
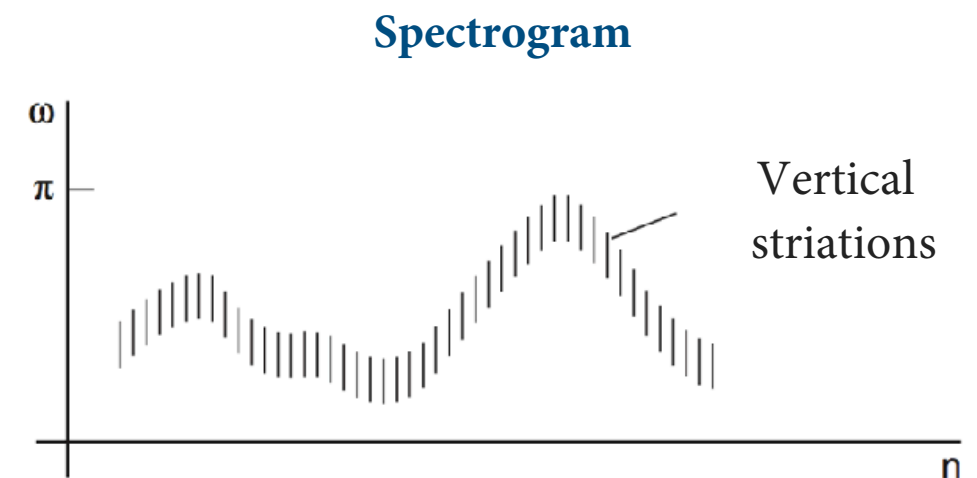
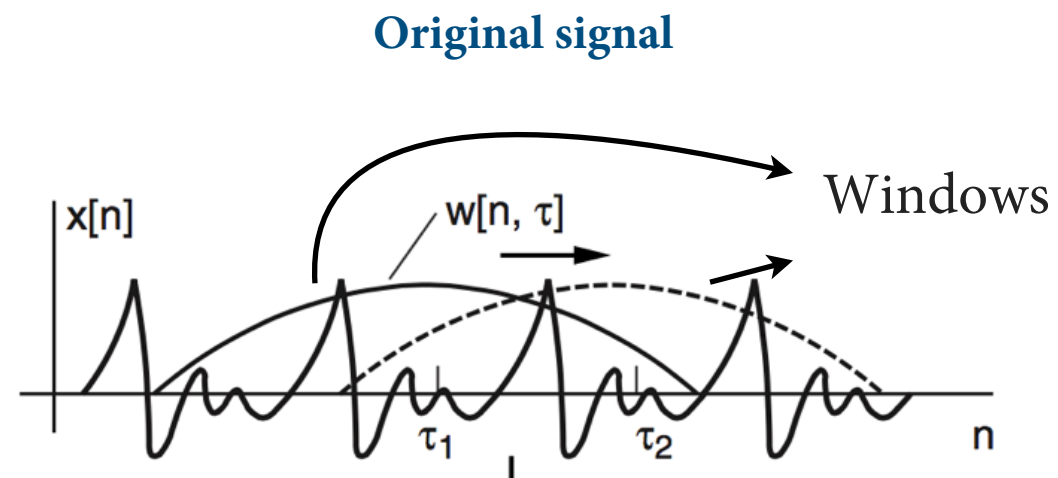
Two « middle C »



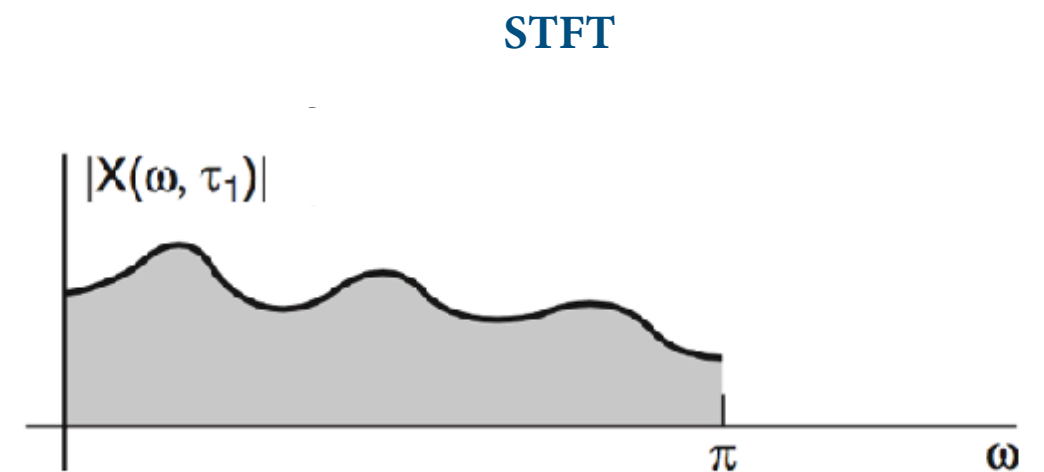
Piano et violin

Spectrogram

- Short temporal window $\Leftrightarrow$ Wideband spectrogram



Smoothing



Spectrogram

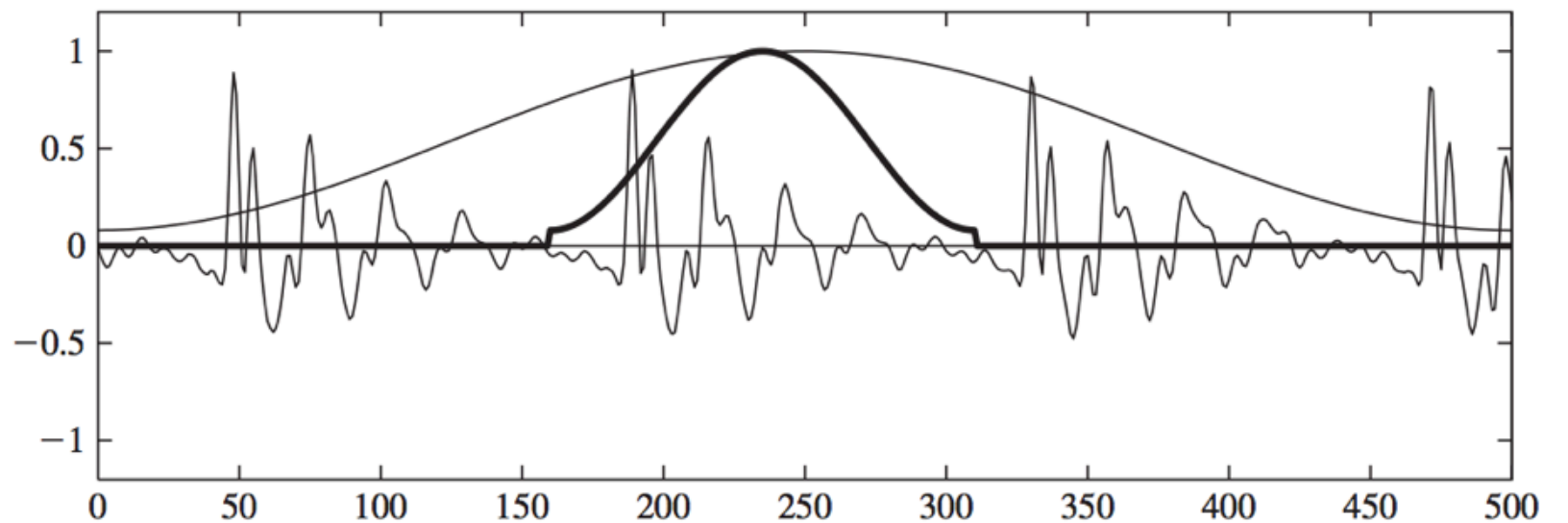
Comparison

Spectrum analysis for voiced sound /AE/ as in /HH AE D/.

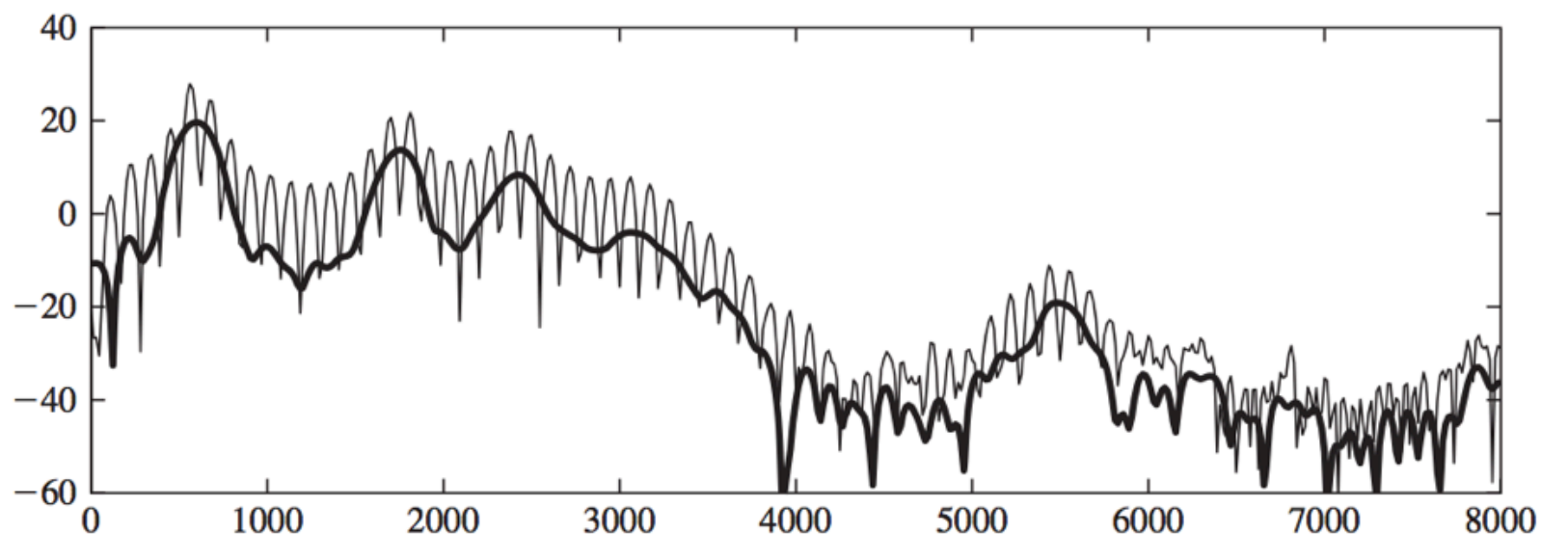
(a) Time waveform (sampling rate 16 kHz) and Hamming windows of length $L = 501$ and $L = 151$ (in bold) samples.

(b) Corresponding spectra.

(a) Voiced Speech with 501- and 151-point (in bold) Hamming Windows



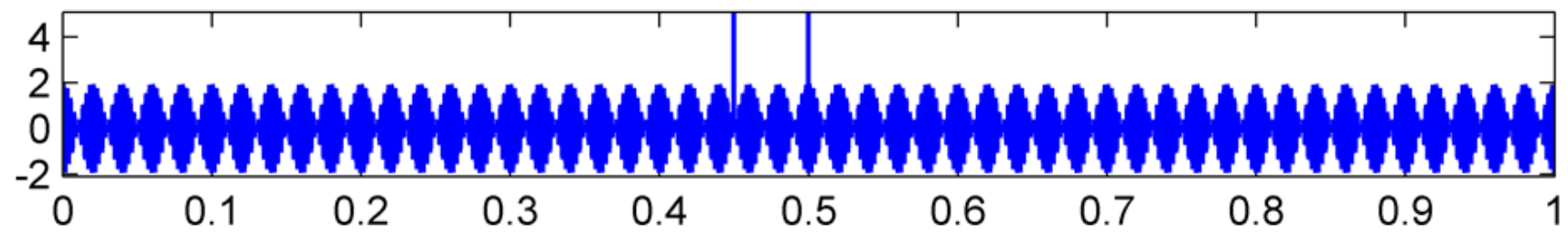
(b) Corresponding Narrowband and Wideband Spectra in dB



Spectrogram

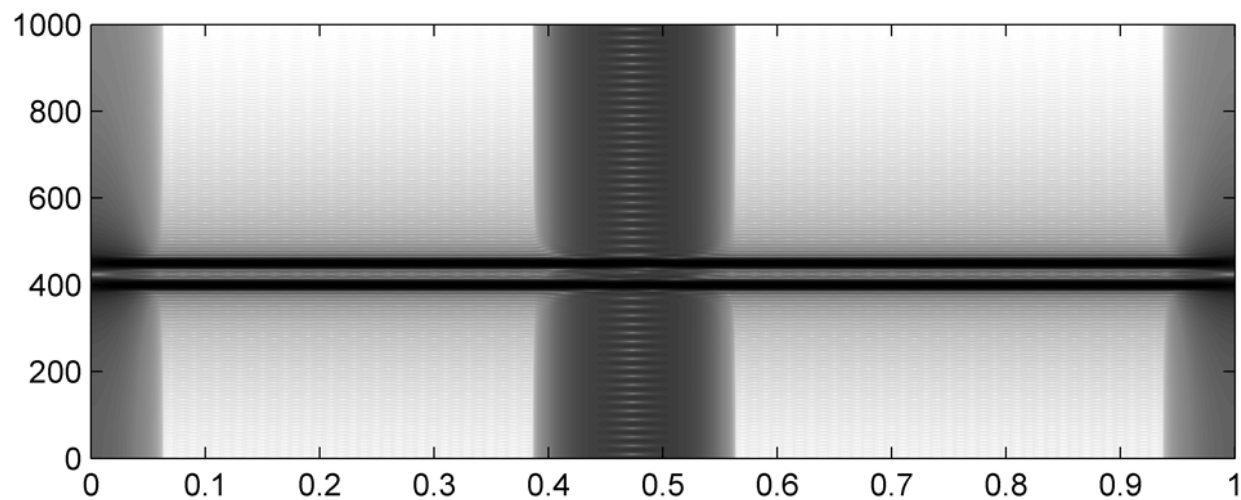
Comparison

Sound waveform



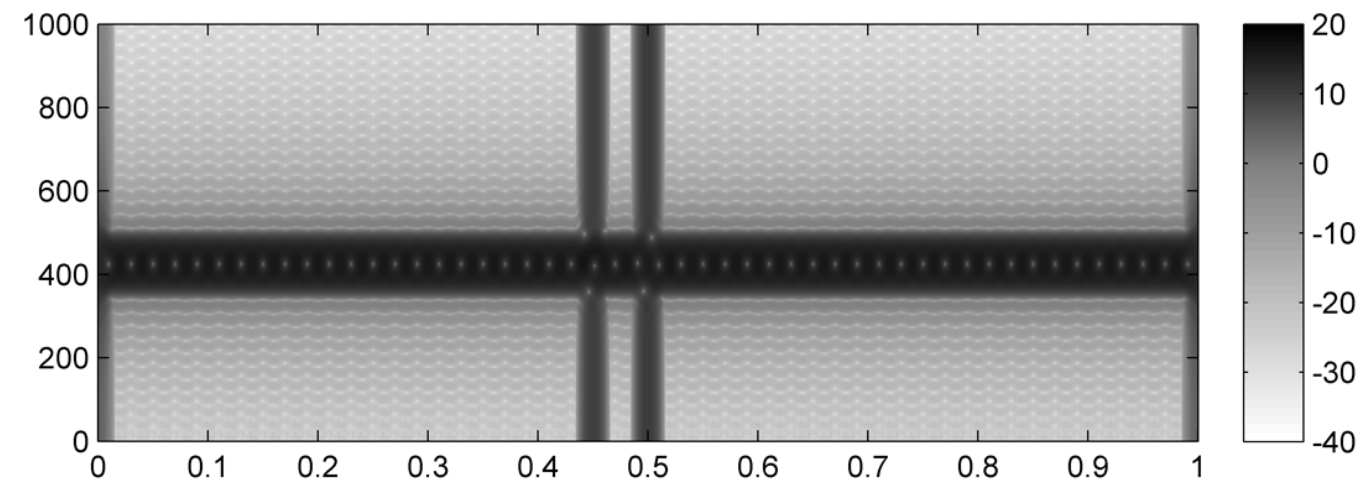
Two sine functions (at 400 and 450 Hz) + two impulses at $t = 0.45$ s and $t = 0.5$ s

Wide window (128 ms)



The impulses are smoothed

Short window (32 ms)



Sine frequencies are smoothed

Speech Spectrogram examples

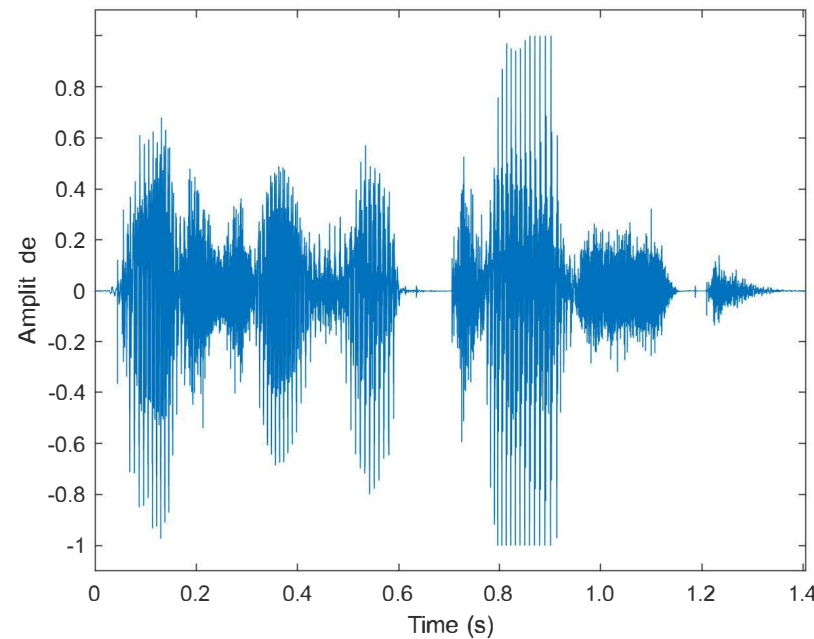
The parameters of the speech and the spectrogram computation for the utterance “This is a test” are as follows:

- Speech parameters
 - Sampling rate (F_s): 16 kHz
 - Speech duration: 1.406 seconds
 - Speaker: male
 - Utterance: “This is a test”
- Wideband spectrogram parameters
 - Analysis window ($w[n]$): Hamming window (default)
 - Analysis window duration (L): 6 msec (96 samples at the 16 kHz rate)
 - Analysis window shift (R): 10 samples
 - Analysis window overlap ($L - R$): 86 samples
 - Number of analysis sections (N_R): 2250
 - Size of FFT (N): 512
 - Dynamic range of spectral log magnitudes: 40 dB
- Narrowband spectrogram parameters
 - Analysis window ($w[n]$): Hamming window (default)
 - Analysis window duration (L): 60 msec (960 samples at the 16 kHz rate)
 - Analysis window shift (R): 96 samples
 - Analysis window overlap ($L - R$): 864 samples
 - Number of analysis sections (N_R): 235
 - Size of FFT (N): 1024
 - Dynamic range of spectral log magnitudes: 40 dB

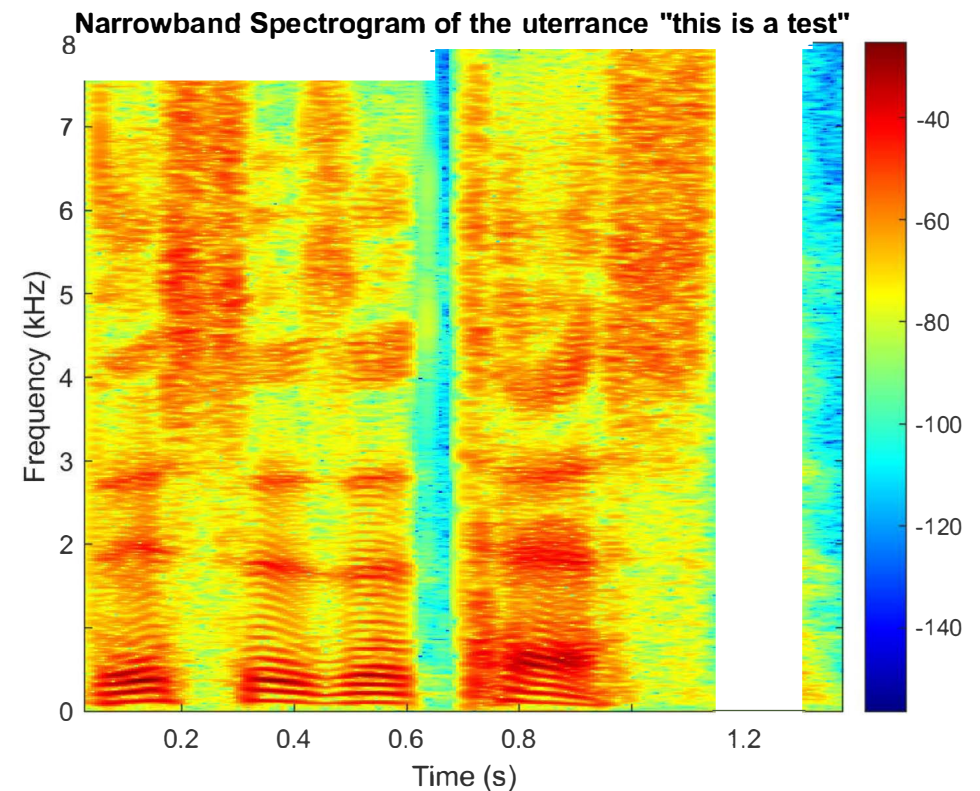
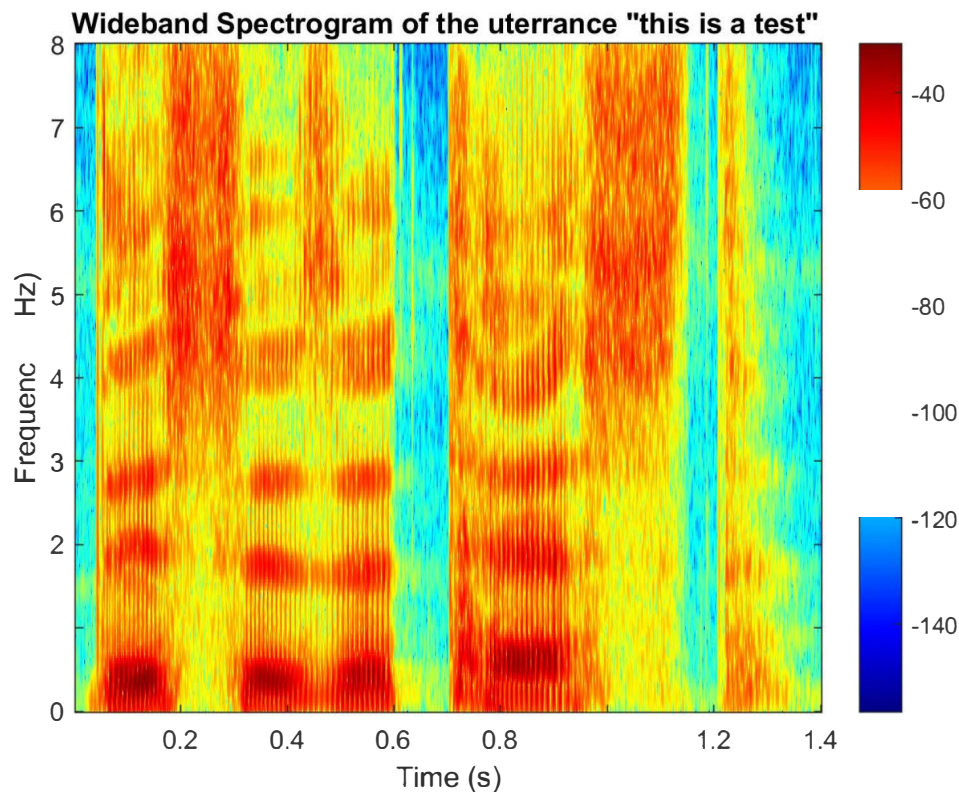
Spectrogram

Example :

Male speaker's
utterance "this is a
test"



The top figure shows the waveform of a male speaker's utterance of the text "this is a test". At the bottom, the left and the right figures show, respectively, the wideband and the narrowband spectrograms, corresponding to the parameters given in the previous slide.

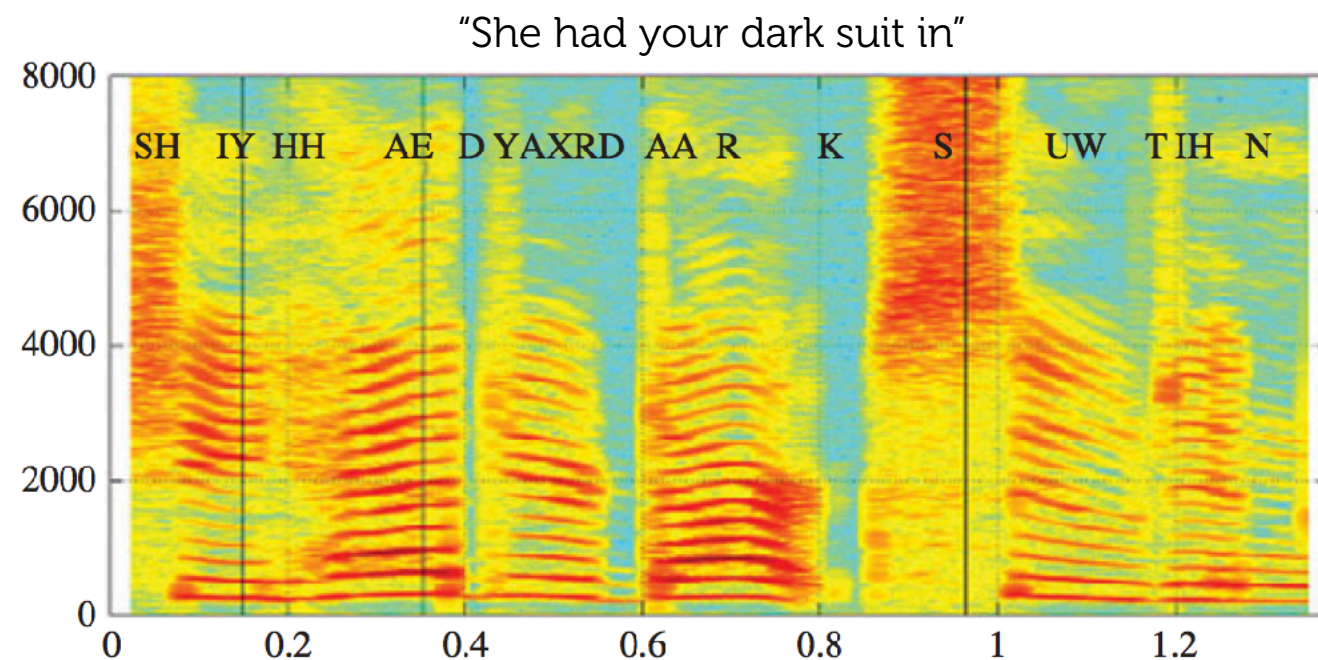


Spectrogram

Example : female speaker

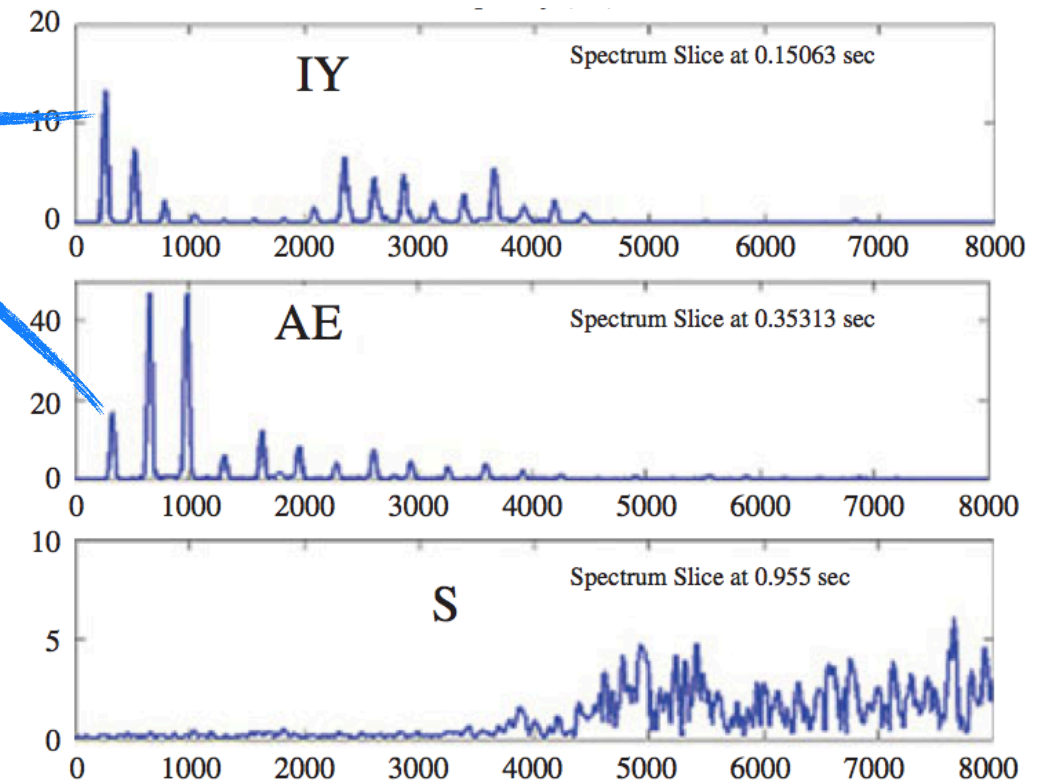
Wide window

Spectrogram



Pitch frequency

f_0



Spectrograms of a female speaker's utterance of the text "She had your dark suit in". The left panel shows narrowband spectrogram using $L = 800$ samples (50 msec) analysis windows, with an 1024-point FFT, and Overlap = 890 samples between adjacent short-time spectra. The right figure shows the narrowband slices at the three time slots indicated by lines in the sound spectrograms, corresponding to the sounds /IY/, /AE/, and /S/.

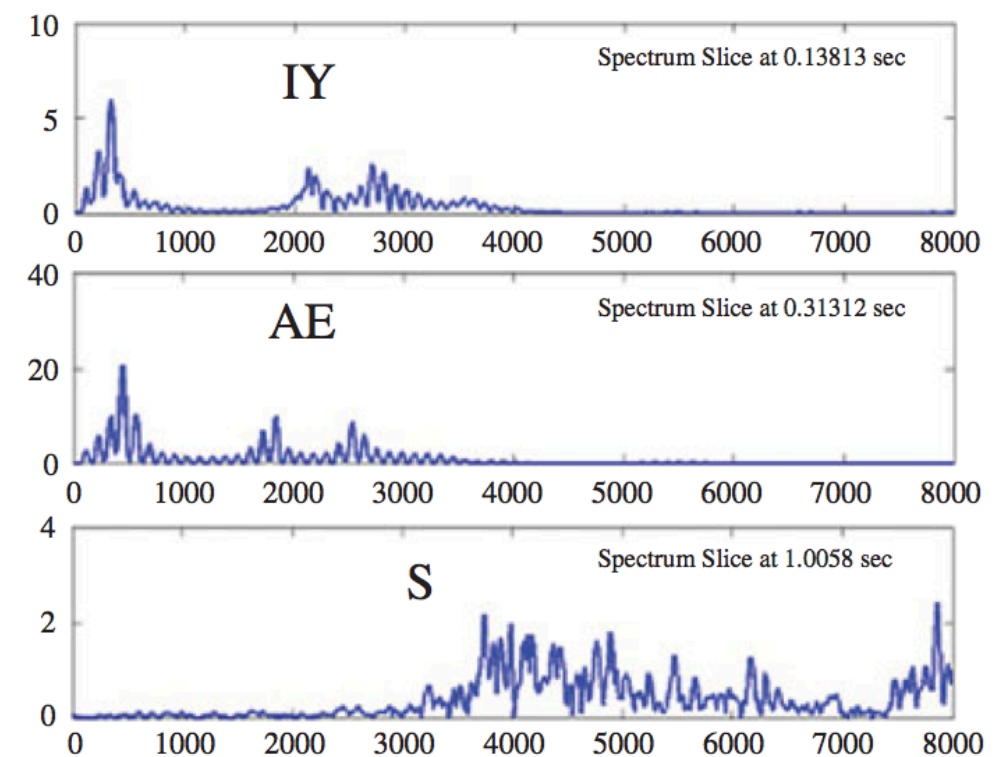
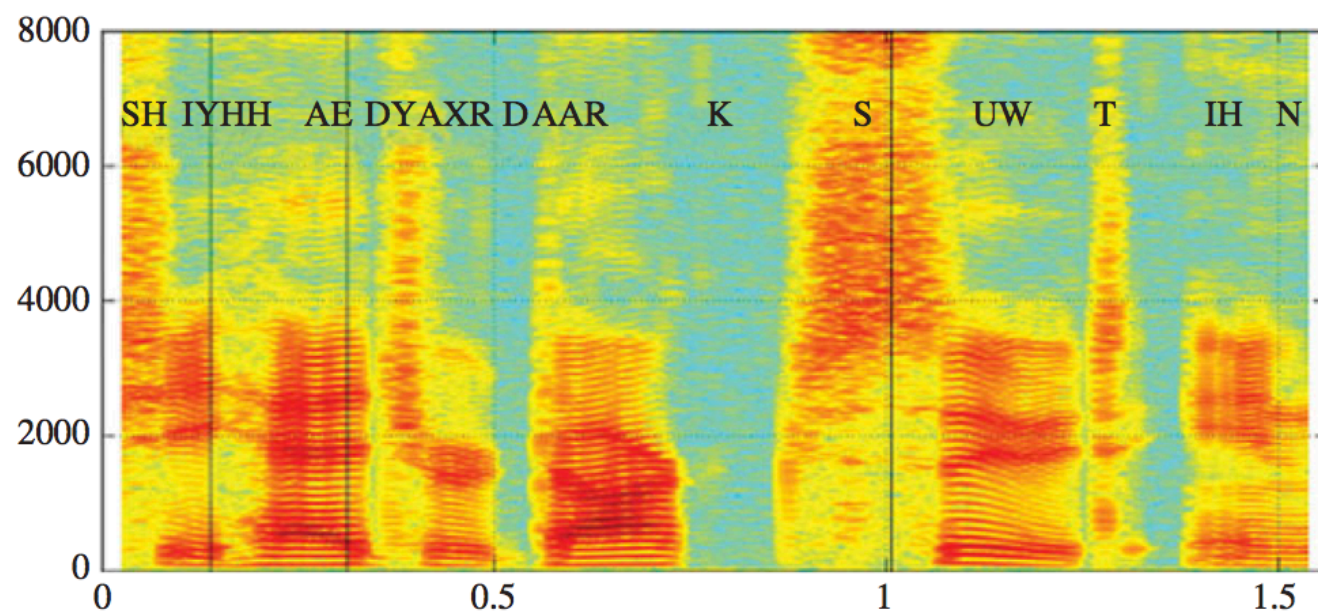
Spectrogram

Example : male speaker

Wide window

Spectrogram

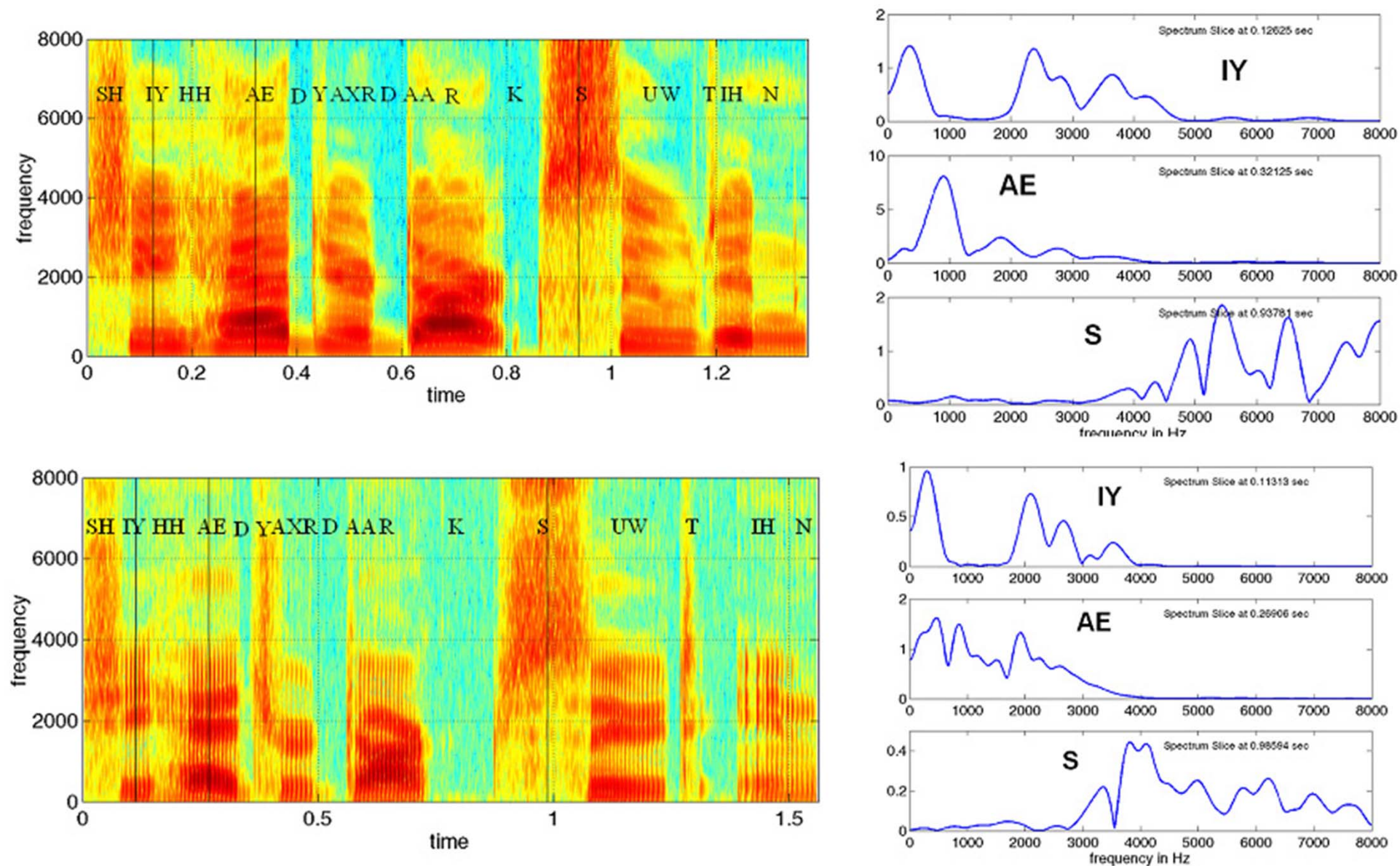
"She had your dark suit in"



Spectrograms of a male speaker's utterance of the text "She had your dark suit in". The left panel shows narrowband spectrogram using $L = 800$ samples (50 msec) analysis windows, with an 1024-point FFT, and Overlap = 790 samples between adjacent short-time spectra. The right figure shows narrowband slices at the three time slots indicated by heavy lines in the sound spectrograms, corresponding to the sounds /IY/, /AE/, and /S/.

Spectrogram

Example : Wideband spectrogram for female and male speakers



Spectrograms of female (top) and male (bottom) speakers utterance of the text "She had your dark suit in." The left figure shows wideband spectrograms using $L = 80$ sample (5 msec) analysis windows, with an 1024-point FFT, and Overlap = 75 samples between adjacent short-time spectra. The right panel shows wideband slices at the three time slots indicated by heavy lines in the sound spectrograms, corresponding to the sounds /IY/, /AE/, and /S/.

Spectrogram

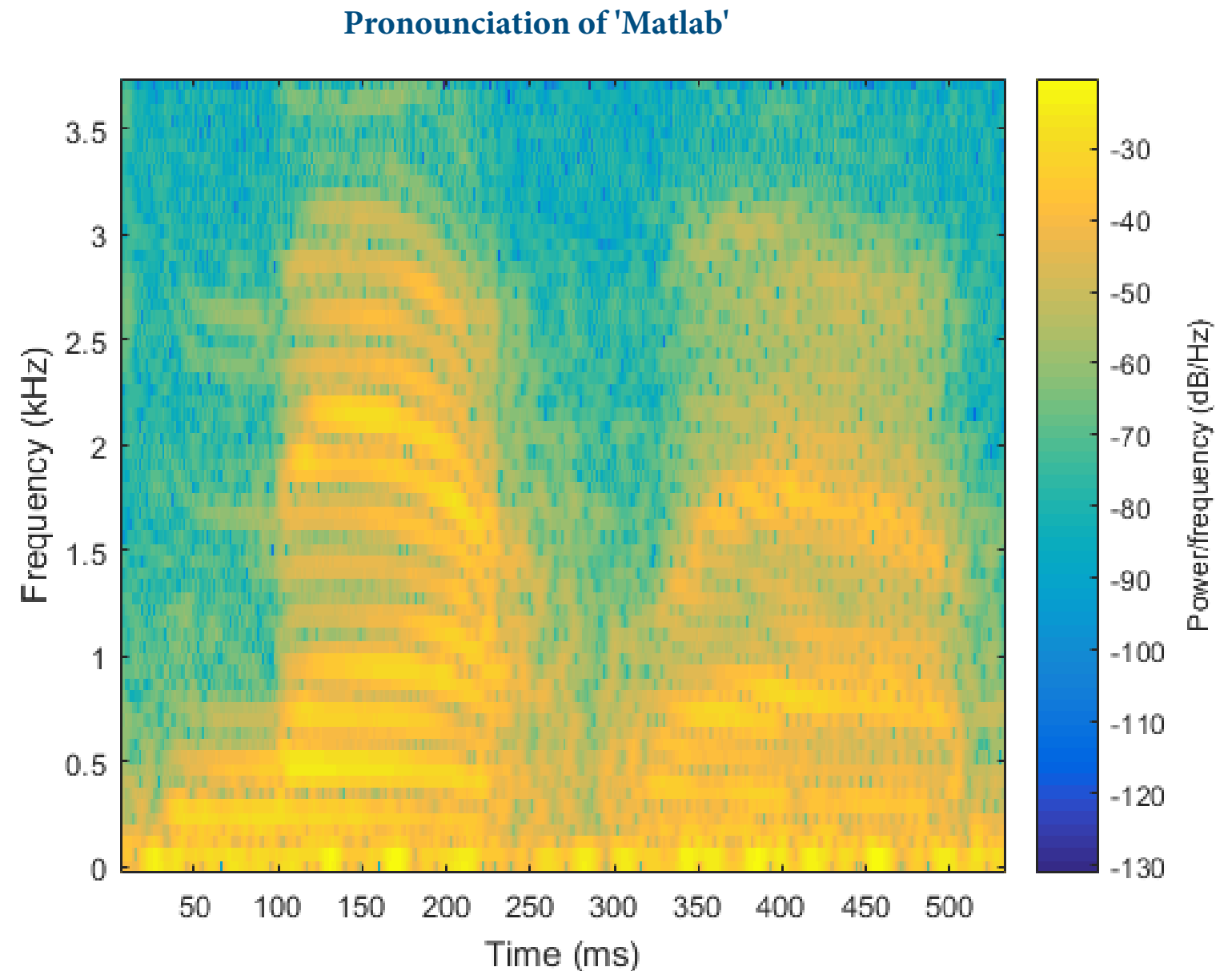
Examples

Changes at 20-40 ms

➔ we target T : 20-40 ms

➔ Overlap : entre 5 - 10 ms

$T = M \times T_s$: Window size in ms



$$f_s = 7418 \text{ Hz}$$

Hamming window of 100 samples

$$T = 100/7418 = .0135 \text{ s}$$

Overlap : 90 samples

Frame size : 128

samples

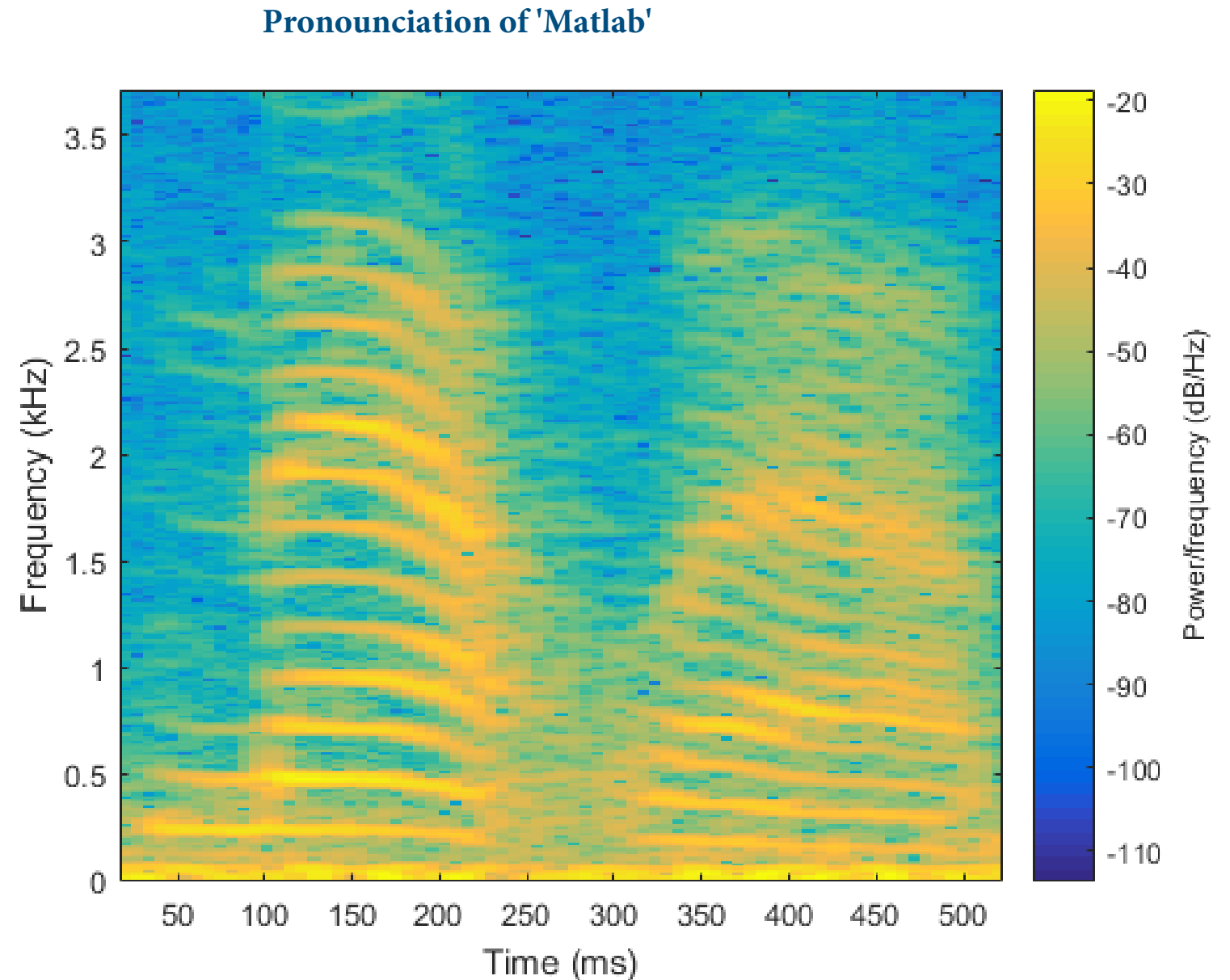
Spectrogram

Examples

Changes at 20-40 ms

➔ we target T : 20-40 ms

➔ Overlap : entre 5 - 10 ms



$$f_s = 7418 \text{ Hz}$$

Hamming window of 300 samples

$$T = 300/7418 = 40.442 \text{ ms}$$

Overlap : 250 samples

Frame size : 512 samples

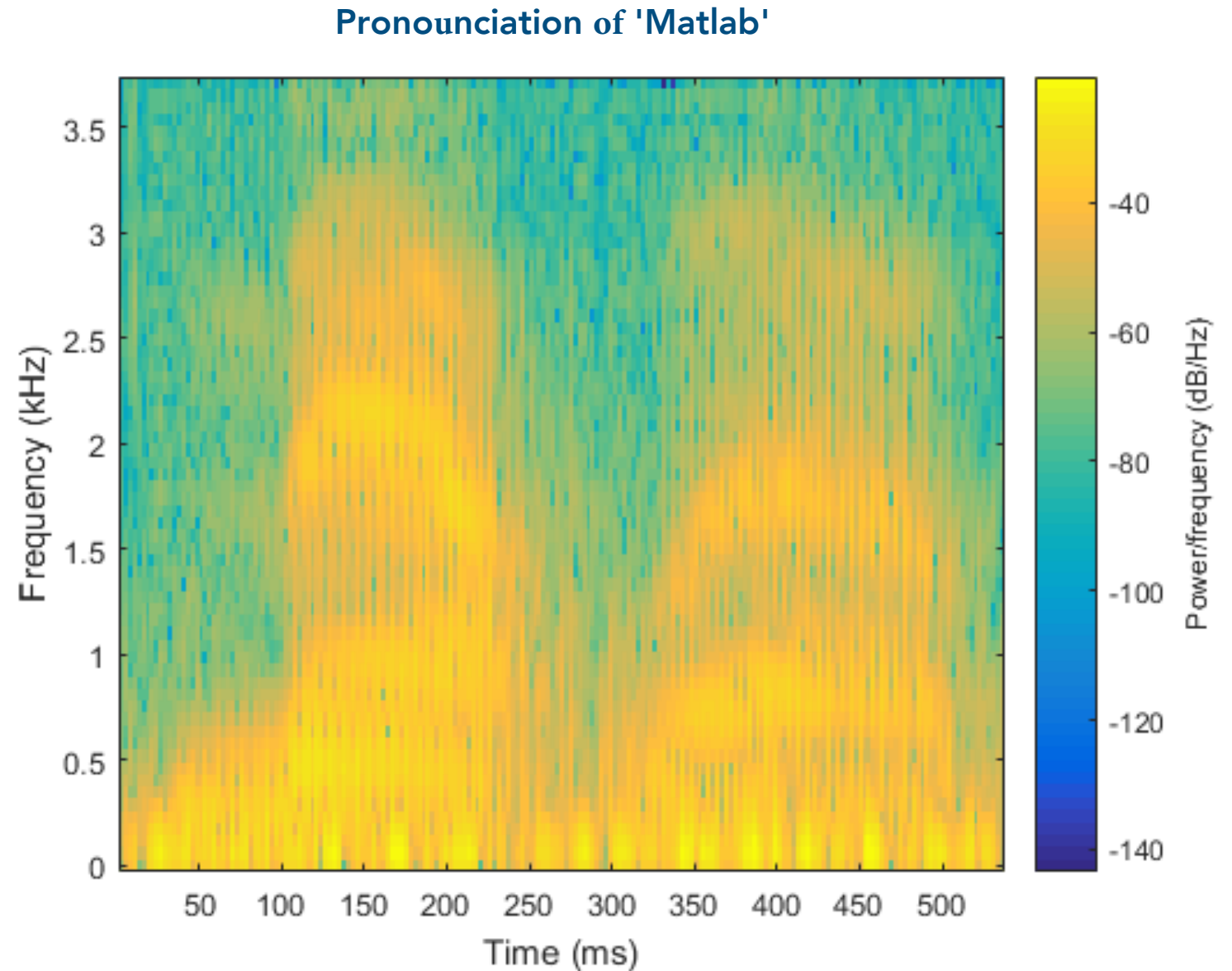
Spectrogram

Examples

Changes at 20-40 ms

➔ we target T : 20-40 ms

➔ Overlap : entre 5 - 10 ms



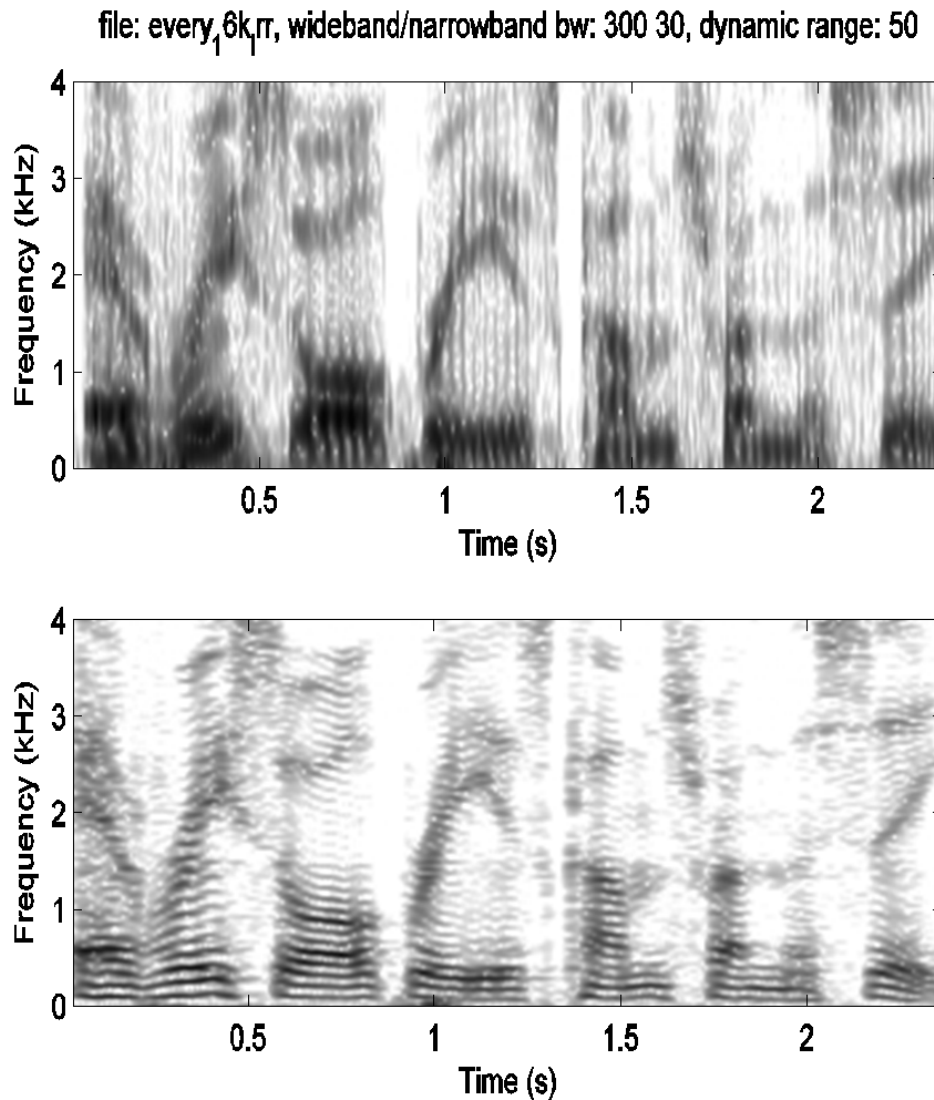
$$f_s = 7418 \text{ Hz}$$

Hamming window of 40 samples

Overlap : 20 samples

Frame size : 128 samples

Wideband and narrowband spectrogram features



➤ Wideband spectrogram

- follows broad spectral peaks (formants) over time
- resolves most individual pitch periods as vertical striations since the IR of the analyzing filter is comparable in duration to a pitch period
- what happens for low pitch males—high pitch females
- for unvoiced speech there are no vertical pitch striations

➤ Narrowband spectrogram

- individual harmonics are resolved in voiced regions
- formant frequencies are still in evidence
- usually can see fundamental frequency
- unvoiced regions show no strong structure